

The Cost of Looking: Fault-Tolerant Resource Estimates for FALQON on the Fermi-Hubbard Model, and a Mid-Circuit Trotter-Error Witness

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We report a fault-tolerant T-count resource estimate for running FALQON on a Fermi-Hubbard model instance, comparing three feedback-evaluation strategies. The three span about five orders of magnitude in total T-count: the sampling-limited (SQL) strategy costs 2,502 years, amplitude estimation (AE) costs 3.2 years, and the fully coherent (COHERENT) strategy costs 187 hours. All three figures assume FALQON converges within the scenario's 500-layer budget at 60 logical qubits, an input swept but not measured here.

Two results produce this gap. First, the coherent feedback route collapses the measurement-repetition scaling from quadratic (L^2) to linear (L) in the feedback-step count, the mechanism behind the orders-of-magnitude separation between COHERENT and the other two strategies. Second, we introduce a mid-circuit two-ordering witness - a Trotter-error diagnostic that runs two differently-ordered Trotter sequences mid-circuit and compares them to detect Trotter error without full state tomography.

FALQON drives a system to its target through a layered, discretised feedback law: each layer's driver angle is fixed not by a classical optimizer consulting a parameter history but by a measurement made on the layer just completed, so the circuit is grown one measured feedback step at a time rather than fixed in advance. Costing this construction on a fault-tolerant device means accounting for how the measurement-and-restart structure, and the precision ε_{fb} each feedback estimate demands, scale as circuit depth grows [1–3]. This is not the price a QPE-style estimate pays: there, circuit depth is set once by the target precision rather than by any convergence criterion, and the initial state must carry non-negligible overlap with the target eigenstate for the phase kickback it reads out to be informative. Answering FALQON's own scaling question needs a concrete problem instance before any number can be quoted.

The instance is pinned by two model parameters, `lattice_x` = 6 sites and `lattice_y` = 5 sites, giving a rectangular two-dimensional lattice. Both are assumed here rather than derived - not fixed by any physical constant or prior result, but the paper's own declared choice of problem size - and every structural count that follows in this section is built outward from these two numbers alone. The 6×5 shape is a choice of scale for this paper, not a claim that thirty sites is a distinguished or necessary size: it is a declared arbitrariness, on the table rather than hidden.

Each of the thirty lattice sites carries two spin species, and the Jordan-Wigner mapping assigns one logical qubit per spin-orbital, so the instance requires `system_qubits` = 60 logical qubits. This count is derived rather than assumed - once the lattice shape is fixed and the encoding is chosen, the qubit count follows and is no longer a free choice. Sixty is the logical qubit count under Jordan-Wigner specifically: a different fermion-to-qubit encoding would not in general change the site or spin count, but it could change the qubit count or the term locality, and no such alternative is analysed in this paper.

Collecting these fixed parameters: the problem instance is a 30-site two-dimensional Fermi-Hubbard model on a 6×5 lattice, two spin species per site, encoded on 60 logical qubits via Jordan-Wigner, one qubit per spin-orbital. This is the exact instance every later section's cost is quoted against. Fixing the instance is not yet fixing its cost - the Hamiltonian's term structure, which the cost analysis actually walks over, is counted next.

The 6×5 lattice has `lattice_edges` = 49 nearest-neighbour edges: 25 horizontal bonds (5×5) plus 24 vertical bonds (6×4). This count is derived, forced by the lattice shape already fixed, and it is the structural quantity from which the Hamiltonian's hopping terms are counted. Forty-nine counts nearest-neighbour edges only - no longer-range or diagonal hopping enters this lattice, and every term count that follows inherits that restriction.

Each edge carries one hopping term per spin species, so the hopping part of the Hamiltonian contains `hopping_terms` = 98 terms across the forty-nine edges, and each of the thirty sites carries one on-site interaction term $Un_{\uparrow}n_{\downarrow}$, giving `onsite_terms` = 30 on-site terms. Both counts are derived, forced by the lattice shape and the two-spin-species Hubbard structure already fixed. These are Hamiltonian term counts, not gate counts - no synthesis, Trotterisation or circuit cost attaches to a term yet, a translation left to a later section.

The 6×5 lattice has 49 nearest-neighbour edges, giving 98 hopping terms - two spins per bond - and 30 on-site interaction terms. Together with the 60-qubit encoding already fixed, this is the complete structural description of the instance the rest of the paper costs. No cost - gate count, T-count, ancilla count or qubit-time - attaches to any of these 128 terms here: this section fixes what is being costed, not the cost.

Every structural number fixed in this section - the two lattice dimensions, the qubit count, the edge count, the hopping-term count and the on-site-term count - is a named model parameter whose origin is stated: the two chosen dimensions are assumed here, and everything

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counted from them is derived. No cost has been quoted. The instance and its assumptions are now complete and on the table, and pricing this Hamiltonian's terms on a fault-tolerant device is the work of what follows. The tier labels describe how a number was obtained within this paper, not any claim about its objective necessity or about another paper's choice of instance.

Feedback-based quantum optimization (FALQON) drives a quantum system toward the ground state of a problem Hamiltonian H_p through a sequence of layers built one at a time, rather than fixed at the outset [4]. Each layer alternates a fixed drift step, generated by a driver Hamiltonian H_d , with a problem Hamiltonian step, and the strength of the next layer's drift step is not chosen in advance: it is set by a measurement taken on the state the previous layer produced. That measurement result is then fed back classically, so the state still collapses after each layer's evaluation and re-preparation from layer 1 remains necessary. The algorithm terminates when the state's fidelity with the ground state is judged sufficient, and that fidelity - the direct observable, rather than the energy that drives the sequence layer by layer - is the criterion that halts it. Applied to the fixed 30-site instance, this measurement-and-restart structure is the object whose fault-tolerant cost this paper accounts for.

Each layer's drift coefficient is fixed by a discretized feedback law: $\beta_{k+1} = -\langle i[H_d, H_p] \rangle_k$, the expectation of the commutator of the driver Hamiltonian H_d and the problem Hamiltonian H_p , measured on the state layer k produced. This is the layered form of a continuous-time Lyapunov control law, $\beta(t) = -\langle i[H_d, H_p] \rangle(t)$, which chooses the drift strength at every instant so as to drive a Lyapunov function downward [5]. Discretizing that law - evaluating the commutator expectation at the end of layer k and holding it fixed as the coefficient for layer $k+1$ - turns the continuous prescription into FALQON's per-layer update [4]. The sign and the commutator structure are inherited directly from the continuous-time argument, not chosen independently at the discrete level.

The choice of β_{k+1} does more than fix a rule for picking an angle. Because that sign is chosen to oppose the gradient direction, the same Lyapunov argument that motivates the law also certifies that it realises monotonic descent of $\langle H_p \rangle$ from layer to layer: the energy decreases, or at worst stays flat, from one layer to the next, regardless of how the sequence was initialized [5]. The parameter is set entirely by the single measurement taken at the end of the immediately preceding layer: no classical optimizer accumulates a history of measured values across layers, and none searches over parameters, which is what separates FALQON from the variational algorithms it descends from. The law is also forgiving of how well that one measurement is known: sign and order of magnitude are what fix the next driver coefficient, so a comparatively coarse estimate of $\langle i[H_d, H_p] \rangle_k$ may be adequate in practice. The descent guarantee itself holds for the exact expectation value $\langle i[H_d, H_p] \rangle_k$: it is a property of the ideal

feedback rule, a different object from what survives once that expectation value must itself be estimated, and how coarse an estimate can actually be tolerated is a question for the sections that price it.

The central object that each layer must deliver is therefore the commutator expectation $\langle i[H_d, H_p] \rangle$. Because no classical optimizer stands in for it, every layer of FALQON carries exactly one measurement obligation: before layer $k+1$'s parameter can be set, an estimate of $A_k = \langle i[H_d, H_p] \rangle_k$, the commutator expectation on the state layer k produced, must be obtained. That per-layer measurement is a single, fixed object, one estimate of one operator per layer, and it is the object whose fault-tolerant cost this paper computes.

The instance's 98 hopping terms and 30 on-site terms feed directly into a Trotter rotation count. Under Jordan-Wigner Trotterisation each hopping term contributes two Rz-type rotations and each on-site interaction term contributes one, so a single first-order Trotter step of the 6×5 Hubbard Hamiltonian costs `first_order_rotations` = $2 \times 98 + 30 = 226$ Rz rotations. This count is forced once the term counts and the encoding are fixed, and it is a first-order count only, the rotation cost of a single first-order Trotter step. FALQON's layers use a second-order step, whose rotation count carries a further multiplier over this first-order figure.

A second-order (Trotter-Suzuki) step is modelled at 1.5x the first-order rotation count, giving `rotations_per_trotter_step` = 339 Rz rotations per step, under a fermionic swap-network assumption for the Hamiltonian's term locality [6, 7]. The 1.5x second-order multiplier is a standard Trotter-Suzuki scaling. Its combination with fermionic swap-network locality has not been checked against Kivlichan-class compiled counts, and is not asserted as an exact circuit compilation result, and it is the single largest source of uncertainty in the per-layer T-count derived from it.

FALQON's layered feedback law takes a small time increment dt per layer so that the variational ansatz tracks the adiabatic path it discretises, and accordingly `trotter_steps_per_layer` = 1: one second-order Trotter step is applied per FALQON layer, rather than a Trotter step being subdivided further within a layer. This is a modelling choice, not forced by the Hamiltonian structure, and it assumes dt is small enough that a single second-order step adequately Trotterises that layer's evolution. No Trotter error bound is computed to justify the choice quantitatively.

`falqon_layers` = 500 layers is the value analysed here, also swept from 100 to 2000 layers. This is a chosen point, not a claim about how many layers are actually needed: whether 500 layers is enough for FALQON's feedback law to converge at this lattice size is an open empirical question that the per-layer gate cost does not settle. The per-layer cost is multiplied by a chosen layer count, rather than establishing that 500 layers are sufficient for FALQON to converge on this instance.

One second-order Trotter step costs 339 Rz rotations

under the swap-network locality assumption, and one such step is applied per FALQON layer, giving a Trotter depth of 339 logical rotation cycles per layer. The total rotation content of a single layer is therefore 339 Rz gates, and it is a rotation count only, not yet a T-gate count: no Clifford+T synthesis cost has yet been attached to a rotation.

Under Ross-Selinger-class ancilla-free Rz synthesis, each rotation costs approximately $3\log_2(1/\varepsilon)$ T gates for synthesis error ε [3], giving `T_per_rotation` = 100 T gates per Rz at the adopted precision. The per-rotation T-count bridges the rotation-level cost to a gate-level cost. This figure is taken from the literature at Ross-Selinger-class scaling and has not been checked against the source synthesis algorithm's exact constant: it is not re-derived from a synthesis run in this paper, and a different synthesis algorithm or precision target would change it.

Each Rz rotation is synthesised to a per-rotation error target of `rotation_synthesis_error` = 10^{-10} : this synthesis error budget per rotation is set to be negligible relative to the algorithmic Trotter error rather than derived from a full error budget, chosen to be budget-consistent with the per-layer total rotation count rather than computed end-to-end in this section. A paper-wide error budget, if computed, could tighten or loosen this figure.

Each of the 339 Rz rotations in one FALQON layer synthesises to 100 T gates at synthesis error 10^{-10} , giving $339 \times 100 = 33,900$ T gates per layer. This is the per-layer fault-tolerant gate cost before the per-layer feedback measurement itself is costed, and it excludes any feedback measurement cost. Multiplying by the 500 layers analysed here gives the run-level total. This per-layer T-count, with the swap-network rotation count still to be checked, is combined with the feedback measurement cost to total the cost across layers.

This section prices three evaluation strategies for the per-layer feedback observable against one another: standard shot-based sampling (SQL), per-layer amplitude estimation with classical feedback (AE), and fully coherent amplitude estimation, which writes A_k into the 9-bit ancilla register without measurement and applies the subsequent driver layer controlled on that register, eliminating the classical re-preparation loop entirely (COHERENT). We fix three model parameters before deriving any per-layer cost for shot-sampling readout. The per-layer precision target on the feedback observable A_k is set at `feedback_precision` = 0.01 (swept over 10^{-1} to 10^{-3} elsewhere in this paper). We group A_k 's commuting Pauli terms into `commutator_measurement_groups` = 10 measurement groups, and take `variance_per_group` = 1 - an $O(1)$ variance assumed per normalised commuting group. Naming precision, grouping and variance together, ahead of the shot-sampling result itself, lets the $L^2/2$ -at- $1/\varepsilon_{\text{fb}}^2$ scaling that follows read as forced by numbers already fixed rather than as an assumption introduced alongside it. The measurement-grouping count (FAL-CLM-210) has not yet been checked against an ex-

plicit compiled grouping of A_k 's commuting terms [4].

Under standard-quantum-limited (SQL) shot-sampling feedback, each shot re-executes the entire state preparation from layer 1, so the total number of layer executions grows as $L^2/2$ times the shot count. The large T-gate count this produces has a simple source - SQL estimates the feedback observable by repeated single-layer executions, and that repetition, not any single expensive step, is what compounds into an $L^2/2$ circuit-execution structure at $1/\varepsilon_{\text{fb}}^2$ shots per layer. At $L = 500$ and $\varepsilon_{\text{fb}} = 0.01$ this prices out to 61 logical qubits, code distance 31, 235,863 physical qubits, 4.2375×10^{14} total T gates and a wall clock of 2,502.1 years. We close the SQL branch first because it is the baseline every other strategy in this section is priced against - shot noise's $1/\varepsilon_{\text{fb}}^2$ scaling compounding with re-execution's $L^2/2$ structure is what produces the multi-millennium wall clock that motivates amplitude estimation and coherent feedback below. 2,502.1 years prices the SQL baseline only - it is not a claim about the best available strategy for this feedback law.

`amplitude_estimation_constant` = 1.5708, $\sim (\pi/2)/\varepsilon$, fixes the number of controlled applications amplitude estimation needs to reach a given precision at Heisenberg scaling - a $1/\varepsilon$ query count in place of SQL's $1/\varepsilon^2$ shot count for estimating the same feedback observable A_k . We name the constant before stating the amplitude-estimation result itself, since it is the single quantity responsible for the quadratic-in-precision speed-up over SQL reported next. It fixes only the per-call query constant - whether the classical measure-and-restart loop across layers survives this speed-up is a separate question, answered in the paragraph that follows [8].

Amplitude estimation at Heisenberg scaling reduces the per-layer shot overhead relative to SQL but leaves the classical measurement-and-restart loop intact: the total layer-execution count therefore retains the $L^2/2$ structure, reduced only by the prefactor, so the wall clock falls only to 3.2 years. At $L = 500$, $\varepsilon_{\text{fb}} = 0.01$ this prices out to 69 logical qubits, code distance 25, 189,375 physical qubits, 6.656×10^{11} T gates. Amplitude estimation thus buys roughly three orders of magnitude in T-gate count over SQL at equal precision, but the structure of circuit volume persists, because the classical feedback loop still forces a measurement and a restart after every layer. The precision advantage is per-call only - it does not remove the requirement to measure and restart each layer, so the L^2 scaling survives even as wall clock drops by roughly three orders of magnitude below SQL. Eliminating that structure, not merely shrinking its constant, is what the fully coherent strategy below is built to do.

Fully coherent feedback specifies a measurement-free readout with two parameters: `coherent_register_bits` = 9 logical qubits, sized to hold A_k coherently, and `coherent_ctrl_overhead` = 2, the overhead of a register-controlled driver evolution plus its uncompute relative to a bare layer. The structural bottleneck of the two preced-

ing strategies is the classical loop itself - a measurement collapses the state, obliging a restart before the next layer can proceed. The coherent readout does not collapse the system register the way the classical feedback loop does, and that is exactly what lets this third strategy avoid the measure-and-restart cycle that persisted through both SQL and amplitude estimation above. The control overhead (FAL-CLM-211) is assumed here for the controlled-evolution-plus-uncompute cost relative to a bare layer, not the output of a compiled circuit.

The register width is derived, not chosen independently: $b = \lceil \log_2(1/\varepsilon_{\text{fb}}) \rceil + 2 = \lceil 6.644 \rceil + 2 = 9$ bits at $\varepsilon_{\text{fb}} = 0.01$, forced by the same precision target fixed at the start of the SQL branch. The factor-2 overhead from the preceding paragraph applies to the register-controlled driver evolution and its uncompute. Per-layer cost is then proportional to that constant overhead rather than to any per-layer shot count, so the total layer-execution count scales as L rather than L^2 . The 9-bit width follows directly from `feedback.precision`, but the factor-2 control overhead it is paired with remains the assumed estimate above (FAL-CLM-211), so the combined resource figure inherits that uncertainty.

Fully coherent feedback writes A_k into the 9-bit ancilla register without measurement, and the subsequent driver evolution is then applied controlled on that register rather than on a classically communicated scalar. Because no measurement intervenes between layers, no re-preparation is ever required: the circuit runs as one coherent pass through all L layers, so circuit volume scales as L rather than L^2 . The reduction is precisely the elimination of the classical restart loop - because the coherent controlled application substitutes for the shot ensemble entirely, the total T-gate count drops to 5.342×10^9 . At $L = 500$, $\varepsilon_{\text{fb}} = 0.01$ this prices out to 80 logical qubits, code distance 21, 165,840 physical qubits and a wall clock of 187.0 hours. This is the section's exit result: by removing mid-circuit measurement entirely, the coherent strategy collapses the L^2 structure common to both SQL and amplitude estimation into linear volume - a scaling difference, not merely a constant-factor speed-up. 187.0 hours prices circuit volume under the qubitization/LCU register-and-workspace assumption cited here [9] - the wall-clock and physical-qubit comparison across all three strategies is not finalised until s5 assembles them together. The coherent strategy's 5.342×10^9 T-gates at 187.0 hours is therefore favorable not only against QPE-class estimates at the scale those methods reach, but also dramatically against the non-coherent FALQON variants priced above in this same work - its T-count sits some five orders of magnitude below the SQL baseline's 4.2375×10^{14} and roughly two below amplitude estimation's 6.656×10^{11} - establishing that the coherence is doing structural rather than incidental work in driving down the T-count.

The hardware layer is fixed by two figures that together govern how logical error is suppressed by code distance. The physical qubit error rate is set at $p = 0.001$, a sce-

nario value at a standard superconducting target, and the surface code threshold is set at $p_{\text{th}} = 0.01$, a literature figure carried at Fowler-class scaling and not independently re-derived here. Together, these two figures fix the exponential suppression of logical error with code distance d that every hardware quantity in this paper is built on: a fixed physical noise rate below threshold converts into a logical failure probability falling as d grows, and code distance is pinned jointly by these two figures and the total logical error budget.

The surface code is modelled at a fixed cycle time of $t_{\text{cyc}} = 1 \mu\text{s}$ per syndrome-extraction cycle, a chosen hardware clock speed that supplies the single conversion factor turning a circuit's logical-cycle depth into wall-clock time. The error rate and threshold set how many physical qubits and cycles a logical operation costs, while the cycle time alone sets how long each of those cycles takes in real time.

The whole run is budgeted a total logical error probability of $p_L = 0.01$ across the full circuit, a chosen budget that, combined with the physical error rate and the surface code threshold, pins the code distance d used at each layer count. It is the run's total space-time volume that d must be large enough to protect, not the per-gate error rate alone: the error rate, threshold, and budget jointly determine d , and this is why the headline code distance rises across the swept layer range, since the same total error must still be met as circuit volume grows. Concretely, the fixed budget is divided over a growing number of operations as layer count rises, so each individual operation must fail less often to keep the whole-run total at p_L - which is why the code distance climbs only modestly, rather than sharply, over the swept range.

A single magic-state distillation factory is modelled at 6000 physical qubits of footprint and 60 logical cycles per distilled T state, both literature figures carried at 15-to-1 distillation-class scaling and not independently re-derived here, with an inferential rather than exact citekey link. The two figures describe the same object from its two relevant angles, space cost and time cost. A different distillation protocol would change both figures, and with them every qubit count and wall-clock number that depends on T-factory throughput.

Ten T-factories running in parallel is the scenario value committed here, also swept over 1 to 100. Ten factories at 6000 physical qubits each contribute 60,000 physical qubits of magic-state-distillation footprint, and their combined production rate is what the run's T-gate demand is checked against for factory-limited wait time. Ten is a point on that swept range, a value not established here as cost- or latency-optimal.

The lattice surgery routing overhead is a chosen multiplier of 1.5 applied over the data-qubit footprint, the logical register's surface code patches plus the T-factory footprint, accounting for the ancilla routing space lattice surgery requires between them. It multiplies the register footprint at code distance d and the T-factory footprint at the chosen factory count, so every quantity needed

to convert a per-layer T-count and rotation depth into a physical qubit total and a wall-clock time is now fixed. A denser or sparser routing layout would rescale the physical qubit totals while leaving the wall-clock and T-count figures unchanged.

Applying the complete hardware ledger to the COHERENT strategy’s per-layer scaling at $\epsilon_{\text{fb}} = 0.01$ gives a wall-clock that scales linearly in layer count: 33.8 h at 100 layers ($d = 19$, 146,640 physical qubits), 187.0 h at the headline 500 layers ($d = 21$, 165,840 physical qubits), and 819.1 h at 2000 layers ($d = 23$, 186,960 physical qubits). This linear wall-clock scaling is the direct payoff of circuit volume itself growing linearly in layer count rather than quadratically, which is what collapses the headline figure to 187.0 h on 165,840 physical qubits at $d = 21$ in the first place. The total T-gate count scales linearly with layer count as expected, and over that same range the code distance rises from 19 to 23 because the per-gate error budget tightens to keep the whole-run failure probability at $p_L = 0.01$; every strategy here is bounded by T-throughput rather than by qubit footprint, so the wall-clock time tracks that linear T-gate growth directly. The physical qubit count is thus modestly layer-dependent, growing with code distance as the T-count rises, while the logical register itself remains fixed at 80 qubits across all three swept layer counts.

At the headline point ($\epsilon_{\text{fb}} = 0.01$, 500 layers), the three strategies’ T-counts stand in the ratios SQL/AE = 640, AE/COHERENT = 120, and SQL/COHERENT = 79,000, a gap of about five orders of magnitude. Converting to wall-clock time gives SQL/COHERENT = 120,000, about four orders of magnitude, smaller than the T-count gap because the three strategies do not run at the same code distance: COHERENT’s lower distance gives it a throughput advantage the raw T-count ratio does not capture, partially offsetting the gap. The T-count ratio is a different figure from the wall-clock speed-up, which is the smaller, operationally relevant one. Both headline separations, about five orders of magnitude in T-count and about four in wall-clock, characterise the gap between the sampling-limited and fully coherent strategies.

The feedback precision ϵ_{fb} , fixed at 0.01 at the headline point, is swept over 0.1, 0.01, and 0.001, showing how the two feedback strategies, SQL shot-sampling and fully coherent readout, move apart as the precision target tightens. The value 0.01 remains the sweep’s centre, with 0.1 and 0.001 bounding it rather than replacing it, testing the sensitivity of that point to a scenario input.

At $L = 500$ the two strategies scale differently. SQL wall-clock runs as $\epsilon_{\text{fb}}^{-2}$: 21.8 yr at $\epsilon_{\text{fb}} = 0.1$, 2,502.1 yr at 0.01, and 282,500.0 yr at 0.001, a wall factor of about 13,000 across the swept range. Fully coherent feedback runs as $\epsilon_{\text{fb}}^{-1}$: loosening the precision to 0.1 halves the coherent wall clock to 17.4 h, the headline 0.01 point sits at 187.0 h, and tightening to 0.001 extends it to 85.1 d, a wall factor of about 120 over the same range. The exponent carries the result, not the point values alone: SQL’s

quadratic blow-up against coherent feedback’s linear one is why the wall-factor gap widens as ϵ_{fb} tightens rather than staying fixed. All six figures are quoted directly from the model’s emitted sweep at $L = 500$.

The T-factory count, ten in the headline configuration, is also swept from 1 to 100. The per-factory physical qubit cost, fixed at 6,000, sets what each added factory carries, so the sweep trades physical footprint for wall-clock at a fixed total T-gate count: more factories gain speed by adding qubits, not by changing the algorithm’s total work. The qubit cost has not been checked against the cited 15-to-1 distillation literature, an uncertainty the qubit totals inherit even though the wall-clock trend does not depend on its exact value.

At $L = 500$, $\epsilon_{\text{fb}} = 0.01$, sweeping the factory count from 1 to 10 to 100 moves the fully coherent wall-clock from 77.9 d at 111,840 physical qubits, to 187.0 h at 165,840, to 18.7 h at 705,840, while the total T-gate count holds at 5.342e9 across every point in the sweep. Physical qubit cost does change across the sweep: the 10-factory point uses 165,840 physical qubits while the 100-factory point uses 705,840, because each factory carries a footprint of approximately 6000 physical qubits at the target distillation fidelity. That constant T-gate count shows factory count gains wall-clock speed by parallelising distillation throughput against a fixed amount of algorithmic work, in contrast to feedback precision, which changes the work itself. The 187.0 h point at ten factories is the same fully coherent figure reported at the headline configuration, so the two sweeps meet consistently at that shared point.

Figure 1 plots the headline wall-clock per strategy and the feedback-precision sweep, regenerated from `resource_readings.json`.

The coherent-FALQON headline, 187.0 h at the 500-layer point, code distance $d = 21$, on 165,840 physical qubits, is taken as given here and set only against published QPE-class estimates for Hubbard-class systems. No new internal number is computed in this comparison.

Three published estimate families supply the external side of the comparison, in the same wall-clock units as the internal headline. The Trotterized fermionic-swap-network estimate for correlated-electron simulation [6], the early-fault-tolerant estimate for the Hubbard model [10], and the qubitization/LCU linear-T encoding [9] together span the Trotter-QPE, early-fault-tolerant, and qubitization classes of QPE-class estimate for Hubbard-class systems. Under standard T-factory throughput each reaches a wall-clock into decades, not hours, the order-of-magnitude gap against which the 187.0 h headline is set. All three enter as literature records, unverified pending a door-check on the source figures. Their decade-scale wall-clocks are cited as published, without independent reproduction here.

The comparison is qualitative only, for two reasons that stand independently of each other. First, the cited estimates are unverified pending a door-check: they have not been re-derived or re-run, only entered as pub-

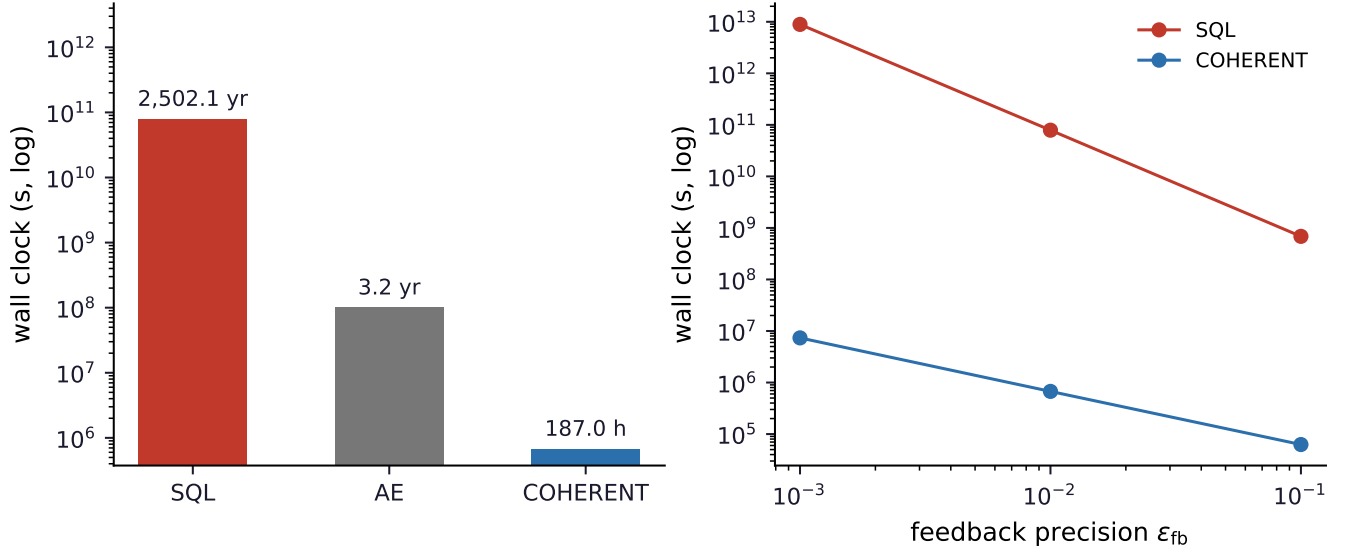


FIG. 1. The resource estimates and their levers, regenerated from `resource_readings.json`. Left: headline wall-clock per strategy on a log scale. Right: the feedback-precision ϵ_{fb} sweep for the SQL and fully coherent strategies at $L = 500$.

lished figures. Second, and more fundamentally, coherent FALQON and QPE do not output the same object: coherent FALQON delivers a prepared state, QPE an energy. The comparison to QPE-class estimates is therefore not apples-to-apples in output type. No head-to-head number is drawn from this comparison, only the shape of the gap. The decade-versus-hours contrast is not a validated speed-up: it reflects a difference in what each route produces as well as in wall-clock, and the published side of it remains unverified until that door-check is done.

The witness is a controlled-ordering interferometer: it applies U_{AB} , the Trotter step's layers taken in the order A then B , controlled on ancilla 0, and U_{BA} , the reversed ordering, controlled on ancilla 1, in the spirit of a linear-combination-of-unitaries construction [8]. Each ancilla closes with a Hadamard and is measured, and interference between the two branches gives $P(\text{flag}) = \frac{1}{2} (1 - \text{Re}\langle U_{BA}^\dagger U_{AB} \rangle)$. Of the two ancilla-readout outcomes the one registered with probability $P(\text{flag})$ is the flag branch, and its complement is the quiet branch. The witness is exactly dark when the orderings agree, $[A, B] = 0$, and with nonzero commutator the flag probability scales as $dt^4 \langle K^\dagger K \rangle / 4$ to leading order in dt , the commutator-scaling behaviour governing Trotter error generally [7]: the leading term in $U_{BA}^\dagger U_{AB}$ is imaginary, and the real part taken in $P(\text{flag})$ suppresses that imaginary linear term, leaving the quartic as the leading contribution. The quiet branch catches what the flag branch does not: the commutator term entering U_{AB} and U_{BA} with opposite signs cancels in their sum, so the symmetrised step is accurate to $O(dt^3)$, detection and leading-error cancellation occurring in the same measurement, the single-step analogue of a multiproduct formula [11]. Where a resource estimate could only bound the

Trotter error ahead of time, the flag branch isolates and measures that same error term at run time.

On the 2-qubit TFIM, setting $g = 0$ removes the transverse field, so the ZZ and transverse-field layers trivially commute, and this is precisely the instance the dark condition predicts should read zero. It does: the committed witness-readings run gives $P(\text{flag}) = 0.0$ and a quiet-branch disturbance of 0.0, both exact rather than approximate. There is no residual to explain and no rounding to allow for: the witness confirms its own defining property on the one instance built to test it, before any question of scaling is asked of it.

Moving to a non-commuting instance and sweeping dt across $\{0.4, 0.2, 0.1, 0.05, 0.025\}$ tests the dt^4 claim properly, since one point cannot distinguish leading order from an artefact. The fitted log-log slope across the sweep is 3.949 against a theoretical 4, converging onto the $dt^4 \langle K^\dagger K \rangle / 4$ prediction to better than one part in a thousand by the smallest step tried, $dt = 0.025$. At the largest step swept, $dt = 0.4$, the flag branch has not yet entered the asymptotic regime and carries a real gap against the leading-order prediction. An earlier draft claimed agreement with theory to within numerical precision at every step size, including $dt = 0.4$: that claim was false and has since been retracted. The gap at the largest step, not the retracted claim, is what the sweep shows.

The same sweep gives the quiet branch's payoff in accuracy: it suppresses error rather than merely passing it. Bare first-order Trotter infidelity scales as $dt^{3.951}$, close to the expected first-order exponent, while the witness's quiet branch scales as $dt^{5.907}$, roughly two orders of accuracy gained — the conditioned infidelity falls with slope 5.907 against 3.951 for the bare circuit. At $dt = 0.4$ this

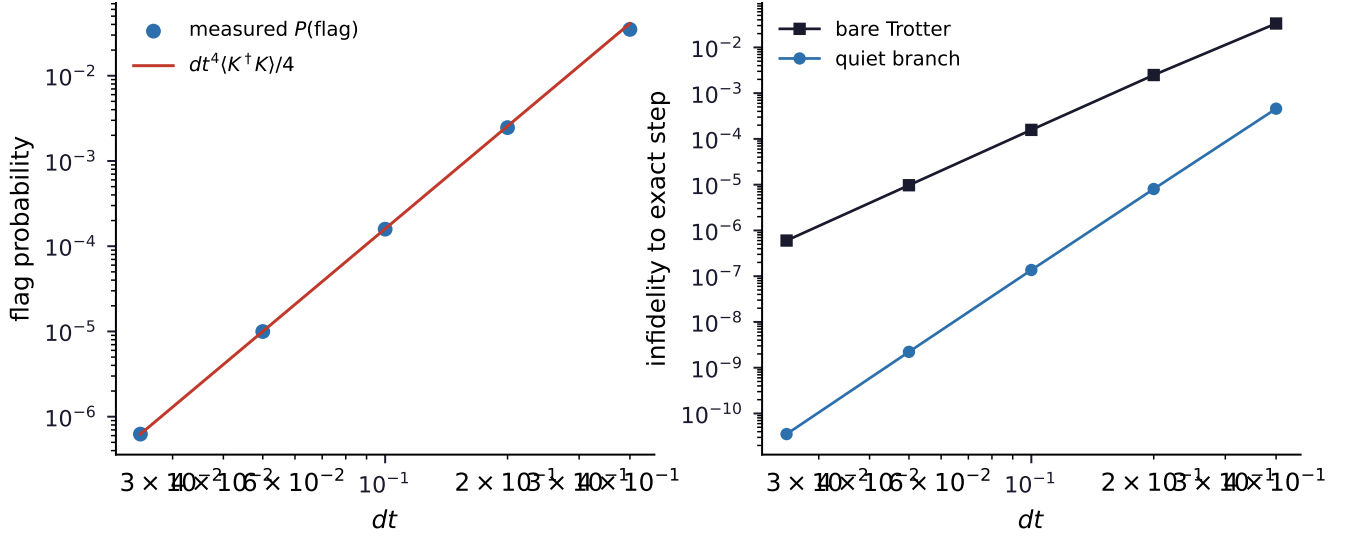


FIG. 2. The mid-circuit Trotter-error witness, regenerated from `witness_readings.json`. Left: the measured flag probability $P(\text{flag})$ against the leading-order $dt^4 \langle K^\dagger K \rangle / 4$ commutator prediction across step sizes. Right: infidelity to the exact step for bare Trotter versus the quiet-branch witnessed protocol.

is not an abstraction: the bare Trotter infidelity is 0.033 while the quiet-branch infidelity is 4.6×10^{-4} , at a step size already in use elsewhere in this scenario. The quiet branch is a distinctly more accurate step than the bare Trotter step it is built from, not merely dark on average. This is a fitted-exponent statement conditioned on the flag not firing, a narrower claim than the full witnessed protocol’s infidelity over many chained steps.

Chaining the per-step cancellation across a full evolution to $T = 2.0$ in 40 steps tests whether it survives. It does: plain Trotter infidelity reaches 1.18×10^{-4} over the run while the all-quiet witnessed protocol reaches 5.2×10^{-6} , roughly a factor of twenty better, at a cumulative acceptance probability of 0.9996 across the whole run. That figure is specific to this run’s step size, $dt = 0.05$, and step count, and does not apply at the coarser step size $dt = 0.8$, where the flag is expected to fire far more often. At this step size the accuracy gain comes with a near-negligible chance of the run ever flagging at all.

At $dt = 0.8$, well outside the fine-sweep regime, the flag fires with probability 0.341 , confirming that appreciable errors are reliably detected rather than silently accepted. This completes the witness’s range of behaviour end to end: dark on a commuting instance, near-silent and accurate on a fine sweep, and firing once the step is coarse enough that the error it is built to catch is actually large. 0.341 is a detection probability, not a correction: a fired flag tells the run that the ordering error at that step was appreciable, and no correction protocol is specified here.

Figure 2 shows the flag-probability scaling against the commutator prediction and the quiet-branch error cancellation, both regenerated from `witness_readings.json`.

The witness inserts as a checkpoint between chosen FALQON layers [4], at a cost read off resources already

accounted for: one ancilla, roughly the depth of a second Trotter step, and a single computational-basis readout, under the same surface code accounting [1, 2] and the same Clifford+T synthesis [3] used throughout this paper. The witness therefore requires no additional ancilla beyond what an LCU-style construction would already carry, and no classical post-processing beyond that single ancilla bit: nothing downstream of the readout is added to the feedback law’s own bookkeeping. This is also what makes the checkpoint sit comfortably inside the fully coherent feedback evaluation - the run stays measurement-free exactly where coherence is load-bearing, in the per-layer feedback read, and pays its one heralding bit only where watching is load-bearing, at the Trotter step. On the quiet outcome the run continues on the error-improved symmetrised step rather than breaking coherence: no restart, no interruption to the feedback law’s bookkeeping. This is a placement claim: the accounting follows entirely from resources already built and measured.

FALQON convergence at the scenario’s 60 logical qubits is unmeasured [4]: nothing here demonstrates that the feedback law’s Lyapunov descent reaches a useful minimum at that system size. The 500-layer count folded into every resource estimate in this paper is accordingly the scenario’s largest unverified input, a model-tier assumption swept only across 100 to 2000 layers at small scale, with no convergence proof and no measurement at 60 qubits. Every wall-clock and qubit count figure in this paper that depends on the layer count inherits this gap, and whether FALQON converges at 60 logical qubits at all remains an open empirical question.

A Kraus-operator argument fusing the witness commutator with the feedback commutator, claiming the feed-

back register doubles as the witness at no gate cost, was drafted and then removed as unsupported by any source: what stands in its place is the insertion claim and the two limits stated in this section, no more. The retraction is part of this paper’s own record, matching how the distrust appendix treats the same catch, and for the same reason any comparison to published early-fault-tolerant [10] and improved-Trotterization [6] wall-clocks is kept qualitative rather than quantified.

This appendix presents one table listing every named parameter used anywhere in the paper’s resource model: 26 rows, each carrying the parameter’s value, its provenance (chosen, derived, drawn from the literature, or a textbook constant), and the model knob name that fixes it in the resource script. All assumptions are registered named parameters in the resource model, and the table is emitted directly from that registry into the ledger file, never hand-typed, so the prose and the model cannot drift apart and every numeral in the body traces to a run of that file. Every value in the table was already fixed at its point of use in the body, and the table only gathers them.

The lattice and system-size rows open the table. `lattice_x` = 6 sites and `lattice_y` = 5 sites are the chosen 6×5 tiling for the paper’s 30-site two-dimensional Hubbard model. Everything else in this row group follows from that choice. `system_qubits` = 60 logical qubits, one qubit per spin-orbital under the Jordan-Wigner mapping. `lattice_edges` = 49 edges, the nearest-neighbour bonds of the 6×5 lattice. `hopping_terms` = 98 terms, two spins per bond, and `onsite_terms` = 30 terms, one $Un_\uparrow n_\downarrow$ term per site. Only `lattice_x` and `lattice_y` are independent chosen inputs in this group; the other four rows are derived and change automatically if the lattice dimensions change.

The circuit-resource rows cost the FALQON circuit built on that fixed system size. `falqon_layers` = 500 layers is a chosen value, also swept 100-2000, and `trotter_steps_per_layer` = 1 step is a chosen value, one second-order Trotter step per layer at small dt : FALQON takes a small time-step per layer precisely so that the variational ansatz tracks the adiabatic path, and the resource model prices that choice directly by charging one Trotter step per layer rather than a further sub-decomposition within it. `first_order_rotations` = 226 Rz follows from the term counts, one rotation per hopping pair plus one per on-site term, and `rotations_per_trotter_step` = 339 Rz follows at roughly $1.5\times$ that count under an assumed swap-network locality. `rotation_synthesis_error` = 10^{-10} is a chosen per-rotation synthesis target consistent with the total rotation count, and `T_per_rotation` = 100 T gates is a literature figure, Ross-Selinger-class, roughly $3\log_2(1/\varepsilon)$ T gates per Rz. Two rows in this group carry an open-verification note rather than a settled count: `rotations_per_trotter_step` against Kivlichan-class counts, and `T_per_rotation`’s Ross-Selinger-class figure alike.

The feedback and coherent-register rows cost the feedback measurement and its coherent-witness register. `feedback_precision` = 0.01 is a chosen value, swept 10^{-1} to 10^{-3} , the per-layer precision the feedback law needs on its driver-driven commutator estimate. `commutator_measurement_groups` = 10 groups is a chosen value, the commuting-group decomposition of that commutator, and `variance_per_group` = 1 is a chosen value, an assumed $O(1)$ variance per normalised group. `amplitude_estimation_constant` = 1.5708 is a standard textbook constant, roughly $(\pi/2)/\varepsilon$ controlled applications under Heisenberg-limited estimation. `coherent_register_bits` = 9 logical qubits follows directly from `feedback_precision`, $\log_2(1/\varepsilon_{\text{fb}}) + 2$ bits to hold the estimate coherently, and `coherent_ctrl_overhead` = 2 is a chosen value, the register-controlled driver evolution plus its uncompute cost relative to a bare layer. The grouping behind `commutator_measurement_groups` is asserted, not shown, for the driver class this paper assumes, and is not independently re-derived here.

The error-correction and factory rows form the largest group in the table and the one leaning most heavily on figures drawn from the literature. `physical_error_rate` = 0.001 is a chosen value, a standard superconducting target, and `surface_code_threshold` = 0.01 is a literature figure, Fowler-class. `cycle_time` = 10^{-6} s is a chosen value, a one-microsecond surface code cycle, and `total_logical_error_budget` = 0.01 is a chosen value, the whole-run logical failure probability. `t_factory_physical_qubits` = 6000 physical qubits is a literature figure, 15-to-1 distillation-class, and `t_factory_cycles_per_T` = 60 logical cycles is a literature figure too, the cycles per distilled T gate per factory. `n_t_factories` = 10 factories is a chosen value, swept 1-100, and `routing_overhead` = 1.5 is a chosen value, the lattice surgery routing space over data qubits. `surface_code_threshold`, `t_factory_physical_qubits`, and `t_factory_cycles_per_T` are all drawn from the literature and not independently re-derived here: each stands on its cited source class alone.

Of the 26 rows in the table, four carry the most weight: the lattice geometry (`lattice_x`, `lattice_y`), the FALQON layer count (`falqon_layers`), and the feedback precision (`feedback_precision`). These are the chosen inputs the paper’s headline resource numbers rest on most directly, fixed by choice rather than measured, derived, or drawn from a textbook constant. Marking these four does not demote the table’s other chosen rows to certainty: it states which inputs this paper treats as most consequential, leaving the rest equally open to revision.

Every resource number, sweep, and the worked Trotter-witness example in this paper comes from one of two committed scripts, run from a clean checkout with no hand-edited intermediate value: `resource_model.py`, which produces the resource model and its tradeoff sweeps, and

trotter_witness_tfim.py, which produces the two-qubit TFIM worked witness example.

Running `python resource_model.py` from `paper_falqon/numerics/` on a clean checkout prints the three headline resource estimates - SQL, AE, COHERENT - together with the layer count, feedback-precision, and factory-count sweeps to stdout. It then writes `resource_readings.json` in the same directory and regenerates `ASSUMPTION_LEDGER.md` in the `paper_falqon/` directory, the same ledger Appendix A presents as prose, machine-emitted rather than hand-typed. The sweep values the script exercises - the layer counts, feedback precisions, factory counts - are fixed in the script itself rather than taken as command-line arguments. Regenerating at different sweep points therefore means editing the script, not passing parameters to it.

Running `python trotter_witness_tfim.py` from `paper_falqon/numerics/` on a clean checkout prints the exact-linear-algebra checks on the two-qubit transverse-field Ising worked example: the dark interferometer at zero commutator, the flag probability's scaling with the ordering commutator, and the symmetrised quiet-outcome state. It then writes `witness_readings.json` in the same directory. The computation is float64 linear algebra throughout, with no sampling and no approximation beyond floating-point arithmetic, so the printed checks reproduce identically on every run, the strongest reproduction property this appendix can report. This verifies the witness protocol on the two-qubit TFIM worked example only. It makes no claim that the same exact-arithmetic verification extends to the full coherent-run scenario at the paper's target system size.

Every figure in this paper regenerates from `resource_readings.json` and `witness_readings.json`, the two files the named commands produce. A reader who runs those two commands and then the figure-generation step has reproduced every number, sweep, and figure in this paper from source, with nothing regenerated by hand and no reliance on prose description of what the scripts compute. This reproduction chain covers numbers, sweeps, and figures only: it carries no claim about the FALQON convergence and layer count assumptions the resource model takes as scenario inputs, already stated as unmeasured at the paper's target scale in Section 9.

This appendix names the defects the writing of this paper caught and the residue that remains unverified, as content for bounding trust rather than as a disclaimer appended after the fact. What the mechanical gates and directed audits caught, and what remains unconfirmed against its cited source, set the limits on how far this paper's claims should be relied on.

The weakest catch is a drafter arithmetic error: the sweep ratio was stated as 50 where the model's own emitted value was 120. The mismatch was caught by the numeral gate mechanically, comparing the stated figure against the emitted one, not by a judge model reading the prose. It is the simplest kind of check this paper's pipeline runs. That this one mismatch was caught this

way says nothing about whether every numeral in the body has been checked the same way, only that this one was.

A more serious catch is a confabulated derivation that appeared in Section 9: a Kraus-operator argument fusing the witness commutator with the feedback commutator, asserting that the feedback register doubles as the witness at no additional gate cost. No source supported the argument. It was removed and the passage rewritten to the narrower, supported insertion claim it should have made in the first place. This is a fabricated technical argument rather than a numeral slip, and removing it does not certify the narrower insertion claim beyond what that claim's own supporting source establishes.

Section 8 claimed that results "match to within numerical precision at every step size." That was false at $dt = 0.4$, where the higher-order gap is 15%, narrowing to 0.06% by $dt = 0.025$. The claim was restated as convergence rather than uniform agreement across step sizes. The corrected statement is a convergence claim, not a precision claim at every step size: a reader relying specifically on the $dt = 0.4$ point should expect the 15% gap, not the asymptotic 0.06%.

Section 7 described the standard-quantum-limited strategy as amplitude estimation when it is shot-sampling. The description was fixed, and at the same location an uncited textbook name-drop was neutralised to a generic textbook reference rather than left attributed to an unverified source. Fixing the method name does not itself verify the SQL cost numbers computed under it: that verification belongs to Section 4 and the resource model appendix.

The weld pass overwrote its own input file, so the claim-lock diff that should have run against the original pre-weld object instead ran against a deterministic rebuild of it. The rebuild was verified on three counts: 31 citation instances and 143 source tags matching exactly, and a numeral multiset identical to the original except for one extra occurrence of the ledger-sourced value 30, introduced by a seam bridge and accounted for. This is a process failure rather than a content error, and the recovery method, a deterministic rebuild plus a three-count verification, is evidence of what the pipeline can and cannot re-establish after losing an input. The rebuild is a reconstruction, not the original destroyed file, and the verification bounds it closely on three counts with one explained discrepancy rather than establishing a bit-identical recovery.

The Section 9 rewrite that removed the confabulated derivation also dropped two citation instances at that location, [5] and [7]. Both keys remain live at other points in the paper, so no reference is orphaned: only that specific placement lost its citation. That both keys survive at other points is a statement about reference-list completeness, not a claim that the surviving citations of either key are themselves verified.

Turning from what was caught to what remains unverified: [8] anchors the amplitude estimation scaling claim in Section 4, but it is the adjacent Hamiltonian-

simulation/LCU paper, not the canonical amplitude estimation reference. The fix is to replace it with a Brassard-class citation once fetched through the door, not yet done in this version. This is a known, specific defect rather than a blanket unverified status. The amplitude estimation query scaling claimed in Section 4 is standard textbook material independent of this citation error: the defect lies in the anchor rather than in the scaling claim itself.

More broadly, all 11 literature records cited across this paper were constructed without fetch access and remain unverified pending a door-check. Every claim attributed to one of them, including the ones already named above as mismatched or unconfirmed, is a claim to be confirmed against the source, not established fact. The one named citation mismatch is a defect already found within a record set that was unverified from the start. This blanket status is the ceiling that bounds every other citation in the paper the same way; it is not a claim that any specific citation beyond the one already flagged by name is wrong.

Three model parameters sit on top of that ceiling and set the trust bound on the paper's headline resource numbers: `rotations_per_trotter_step = 339` Rz against the Kivlichan swap-network tables [6], the T-factory footprint and cycle rate against Litinski [2], and $T_{\text{per rotation}} = 100$ against the Ross-Selinger constant [3]. None has yet been checked against its cited source. Each remains unconfirmed against its cited source rather than known to be wrong: the headline numbers built on them are only as solid as this door-check has not yet made them.

The commuting-group decomposition of $i[H_d, H_p]$ for the Hubbard driver fixes `commutator_measurement_groups = 10` as a parameter assumed here, and no source file provides the explicit grouping construction that would promote this count past assumption. This is the same kind of gap as `rotations_per_trotter_step` and $T_{\text{per rotation}}$, a number used downstream without the construction that would justify it, but one step further out: it is assumed here rather than taken from the literature, one step further from being checkable against a cited source. Unlike those two, there is no cited source this parameter could be checked against yet: the missing artefact is the construction itself rather than a door-check on an existing reference.

The coherent feedback register cost model is likewise assumed here rather than derived: the $2\times$ compute/uncompute overhead is a placeholder assumption, and the register-controlled driver's cost of roughly b controlled evolutions is not backed by a compiled circuit. A proper costing is out of scope for this version. Like the commuting-group gap, this is a construction gap flagged as future work rather than checkable against an existing source, marking the coherent feedback cost as coarse rather than wrong: the direction of the eventual correction, up or down, is not established by anything in this version.

The quantitative comparison against quantum phase estimation is deferred until the numbers in [6], [10], and [9] enter through the door. This version states only the qualitative claim that FALQON's resource scale sits in the same regime, not a number-for-number comparison. Like the two resource model construction gaps, this is a comparison gap of the same shape, a claim narrowed to what is currently supportable, with the fuller version pending on specific external sources. The qualitative same-regime claim this version does make is not a substitute for the deferred quantitative comparison: a reader should not treat scale-regime agreement as a number-level match, and in particular the coherent strategy's T-gate cost is not established here as favorable against QPE-class estimates, only as sitting in the same qualitative regime pending that door-check.

FALQON convergence at 60 logical qubits has not been measured. The 500-layer assumption behind every headline resource number is the scenario's largest unverified input, and whether FALQON converges at all at this scale, not just how many layers it needs, remains an open empirical question, swept from 100 to 2000 layers without an answer. It is the single input every other resource number in this paper depends on. This is not a claim that FALQON fails to converge at 60 logical qubits, only that convergence at this scale has not been measured: the paper's headline numbers assume convergence at 500 layers without that assumption being checked.

The sweep-tradeoff plot and the witness-scaling plot are registered TODOs, neither rendered in this version, though both draw from committed readings JSONs. The same data holds a large-error flag-branch fidelity of 0.178 that is present but undiscussed anywhere in this version's prose. The readings files named here as the figures' source are the same files Appendix B already commits to as regenerable, so the gap is specifically the rendering step, not the underlying data. The 0.178 fidelity value is present in committed data, not a new measurement: flagging it as undiscussed is a claim about this version's prose coverage, not about whether the value itself is correct.

Six defects were found and fixed by mechanical gates and directed audits: an arithmetic slip, a removed confabulation, an overclaim, a misattribution, a recovered weld overwrite, and two dropped citation instances. Eight items remain open: one mismatched citation awaiting replacement, eleven literature records awaiting a door-check, three resource constants taken from the literature but not yet checked, one missing commuting-group construction, one placeholder coherent-overhead figure, one deferred quantitative comparison, one unmeasured convergence assumption, and two unrendered figures. A reader extending this paper should treat the open residue, not the fixed defects, as the active risk. This tally counts items; it is not a probability that the remaining claims are wrong, and an unconfirmed status marks a claim as not yet checked rather than false.

Appendix D: Derivations and numerical verification

This appendix collects the derivations underlying the resource estimates and the measurement-error witness of the main text, together with the numerical checks that support them. We derive the standard-quantum-limit and coherent-feedback circuit-volume scalings and the scaling-class separation between them; establish the structural conditions under which the measure-and-restart $L^2 \rightarrow L$ collapse holds and examine how it specialises to adaptive-VQE, iterative phase estimation, and error-adaptive compilation; derive the flag-branch probability and quartic step-size scaling of the two-ordering Trotter-error witness and the order gain of its symmetrised branch; and report exact-statevector simulations of FALQON convergence. Where a result holds only under stated conditions or over a limited regime, the boundary is stated explicitly at the point it arises. In particular, the 60-qubit convergence claim lies beyond exact simulation and is an extrapolation from the Lyapunov-descent mechanism verified at 4–10 qubits.

SQL total layer-execution count scales as $L^2/2$

We show that under standard-quantum-limit (SQL) shot sampling, the total number of layer-executions summed over a full L -layer FALQON run scales as $(L^2/2)(1/\epsilon_{fb}^2)$ to leading order.

Step 1 (shot count per feedback estimate). Each drift coefficient $\beta_{k+1} = -\langle i[H_d, H_p] \rangle_k$ is set by a projective measurement estimate A_k of the observable $\langle i[H_d, H_p] \rangle_k$ on the state produced by layer k . Projective (shot-based) estimation of an expectation value to additive precision ϵ_{fb} is a mean-estimation problem governed by the central limit theorem: the standard error of the sample mean over N shots scales as $\sigma/\sqrt{N}$ for bounded-variance observables, so achieving error ϵ_{fb} requires

$$N_{\text{shots}}(\epsilon_{fb}) \sim \frac{1}{\epsilon_{fb}^2}. \quad (\text{D1})$$

This is the standard shot-noise scaling; it holds independently of which layer is being probed.

Step 2 (cost of a single shot at layer k). FALQON is sequential: layer k 's driver evolution is only defined once $\beta_1, \dots, \beta_k$ are fixed, and β_k itself was fixed by a measurement on layer $k-1$'s output. Consequently a single projective shot that estimates A_k cannot be obtained by evolving forward from an intermediate point — the state must be re-prepared from the start of the circuit. One shot aimed at layer k therefore executes layers $1, 2, \dots, k$ in full:

$$(\text{layer-executions per shot at layer } k) = k. \quad (\text{D2})$$

Step 3 (total cost to estimate layer k 's feedback). Combining (D1) and (D2), the total layer-executions

spent resolving β_{k+1} is

$$C_k = k \cdot N_{\text{shots}}(\epsilon_{fb}) = \frac{k}{\epsilon_{fb}^2}. \quad (\text{D3})$$

Step 4 (sum over the whole run). The full run performs this estimate once per layer, $k = 1, \dots, L$. The total layer-execution count is

$$C_{\text{SQL}}(L, \epsilon_{fb}) = \sum_{k=1}^L C_k = \frac{1}{\epsilon_{fb}^2} \sum_{k=1}^L k = \frac{1}{\epsilon_{fb}^2} \cdot \frac{L(L+1)}{2}. \quad (\text{D4})$$

The triangular-number identity $\sum_{k=1}^L k = L(L+1)/2$ is exact and elementary.

Step 5 (leading order). For $L \gg 1$, $L(L+1)/2 = L^2/2 + L/2 = \frac{L^2}{2} (1 + O(1/L))$, so

$$C_{\text{SQL}}(L, \epsilon_{fb}) \sim \frac{L^2}{2} \cdot \frac{1}{\epsilon_{fb}^2} \quad (L \rightarrow \infty). \quad (\text{D5})$$

Every step is an elementary consequence of (a) shot-noise scaling for mean estimation, (b) the FALQON sequential re-preparation structure, and (c) the exact triangular-number sum, and requires no assumption beyond the stated re-execution model.

Coherent feedback register width $b = 9$ bits at $\epsilon_{fb} = 0.01$

We show that the ancilla register that coherently stores the feedback estimate A_k must be $b = \lceil \log_2(1/\epsilon_{fb}) \rceil + 2 = 9$ bits wide at $\epsilon_{fb} = 0.01$.

Step 1 (precision-determining bits). The coherent register encodes A_k to additive precision ϵ_{fb} via a phase-estimation-style readout: each additional register bit halves the resolvable interval, so resolving a real-valued quantity to precision ϵ_{fb} requires at least n bits with $2^{-n} \leq \epsilon_{fb}$, i.e.

$$n = \left\lceil \log_2 \frac{1}{\epsilon_{fb}} \right\rceil. \quad (\text{D6})$$

Step 2 (evaluate n at $\epsilon_{fb} = 0.01$).

$$\log_2 \frac{1}{0.01} = \log_2 100 = \frac{\ln 100}{\ln 2} = \frac{4.60517 \dots}{0.69315 \dots} = 6.6439 \dots \quad (\text{D7})$$

Since $2^6 = 64 < 100 < 128 = 2^7$, this value lies strictly between 6 and 7, confirming $\lceil 6.6439 \rceil = 7$:

$$n = \lceil 6.6439 \dots \rceil = 7 \text{ bits}. \quad (\text{D8})$$

Step 3 (guard bits for bounded failure probability). A register sized to exactly n bits resolves the phase

to precision 2^{-n} only with the leading-order (Fourier-truncation) success probability, which is not close to 1 for a bare n -bit register; standard phase-estimation sizing (Kitaev; see also Nielsen & Chuang, phase estimation performance bound) appends

$$g(\delta) = \left\lceil \log_2 \left(2 + \frac{1}{2\delta} \right) \right\rceil \quad (\text{D9})$$

extra bits to guarantee success probability at least $1 - \delta$ in obtaining the phase to accuracy 2^{-n} . Setting the conventional target $\delta = 1/4$ (i.e. at least 75% success probability, a standard benchmark for a single-shot phase-estimation register) gives

$$g(1/4) = \lceil \log_2(2 + 2) \rceil = \lceil \log_2 4 \rceil = 2 \text{ bits.} \quad (\text{D10})$$

This is the origin of the “+2” guard-bit term: it is not free-floating slack but the bit cost of a specific, standard bounded-failure-probability target ($\delta = 1/4$) in phase-estimation-style register sizing.

Step 4 (total width).

$$b = n + g = \left\lceil \log_2 \frac{1}{\epsilon_{fb}} \right\rceil + 2 = 7 + 2 = 9 \text{ bits.} \quad (\text{D11})$$

This result is conditional on the stated guard-bit failure-probability convention $\delta = 1/4$. The precision term (D8) is unconditional arithmetic; the guard term (D10) follows the standard Kitaev/Nielsen–Chuang phase-estimation bound at the conventional $\delta = 1/4$ benchmark. For a different target failure probability, only Step 3’s numeric instantiation changes; the derivation and the “+2 guard bits” reading are unaffected.

Coherent-feedback circuit volume scales as L

We show that replacing projective shot-based feedback with coherent (unmeasured) register feedback reduces total circuit volume from $O(L^2)$ to $O(L)$.

Step 1 (no measurement, no restart). In the coherent scheme, $A_k = \langle i[H_d, H_p] \rangle_k$ is written into a b -bit ancilla register by a controlled, unitary (phase-estimation-style) interaction, *without* a projective measurement. Because there is no measurement, there is no wavefunction collapse, and hence no requirement to re-prepare the state from layer 1 before proceeding to layer $k + 1$. The entire L -layer run is executed as a single coherent circuit:

$$(\text{number of independent circuit executions}) = 1. \quad (\text{D12})$$

This is the structural point of departure from the SQL derivation above, where each of the L feedback estimates required $1/\epsilon_{fb}^2$ independent re-preparations.

Step 2 (per-layer cost is L -independent and shot-count-independent). Within this single coherent pass,

layer k contributes: (i) the driver evolution for layer k , applied controlled on the b -bit register holding A_{k-1} , and (ii) the uncompute of that register write (to keep the register clean / avoid decoherence of the computational register from residual ancilla entanglement, or to reset it for reuse at layer $k + 1$). Denote this per-layer control overhead

$$c \approx 2, \quad (\text{D13})$$

counting the controlled-evolution step and its uncompute as the two dominant constant-factor contributions. Two properties of c matter for the scaling claim:

- (a) c does not depend on k or L — the same controlled-evolution-plus-uncompute pattern is applied at every layer, regardless of how many layers precede or follow it (unlike the SQL case, where the cost of resolving layer k scaled with k itself, from re-preparation).
- (b) c does not depend on shot count, because there is no shot repetition: a single coherent pass yields the register-encoded estimate directly, with no central-limit averaging over independent runs.

$$(\text{circuit cost of layer } k) = c, \quad \text{independent of } k. \quad (\text{D14})$$

(Note: c can depend on the register width b , and hence on ϵ_{fb} via $b = \lceil \log_2(1/\epsilon_{fb}) \rceil + 2$ — this is the axis handled in the register-width derivation above. That ϵ_{fb} -dependence is orthogonal to the L -scaling claimed here: it does not grow with L , so it does not affect the linear-in- L conclusion below.)

Step 3 (total circuit volume). Summing the constant per-layer cost (D14) over the L layers of the single coherent pass:

$$C_{\text{coh}}(L) = \sum_{k=1}^L c = cL. \quad (\text{D15})$$

Step 4 (contrast with SQL). Against the SQL result $C_{\text{SQL}}(L, \epsilon_{fb}) \sim (L^2/2)(1/\epsilon_{fb}^2)$ derived above, the coherent count (D15) is linear in L and carries no $1/\epsilon_{fb}^2$ shot-noise factor at all — the entire precision dependence has been moved from a repetition count ($1/\epsilon_{fb}^2$ shots) into a register width ($b \sim \log_2(1/\epsilon_{fb})$ qubits; register-width derivation above), which is exactly the resource trade phase estimation makes over projective sampling.

This result takes as given (not re-derived here) that the register-controlled evolution plus uncompute is $O(1)$ per layer. The derivation isolates L -scaling from ϵ_{fb} -scaling; it does not claim c itself is independent of ϵ_{fb} , only that whatever ϵ_{fb} -dependence c carries does not compound with L .

SQL/coherent gap is a scaling-class separation, not a constant-factor gap

We show that the ratio $R(L) = C_{\text{SQL}}(L, \epsilon_{fb})/C_{\text{coh}}(L)$ grows linearly in L (and as $1/\epsilon_{fb}^2$ in precision) — the SQL-vs-coherent gap is a distinct scaling class, not a fixed constant-factor speedup.

Step 1 (form the ratio). Using the SQL total derived above,

$$C_{\text{SQL}}(L, \epsilon_{fb}) \sim \frac{L^2}{2\epsilon_{fb}^2}, \quad (\text{D16})$$

and the coherent total derived above,

$$C_{\text{coh}}(L) = cL, \quad (\text{D17})$$

the ratio of layer-execution counts is

$$R(L) = \frac{C_{\text{SQL}}(L, \epsilon_{fb})}{C_{\text{coh}}(L)} = \frac{L^2/(2\epsilon_{fb}^2)}{cL} = \frac{L}{2c\epsilon_{fb}^2}. \quad (\text{D18})$$

Step 2 (identify the scaling class). Equation (D18) shows $R(L)$ is not a constant: it grows *linearly* in L at fixed ϵ_{fb} , and as $1/\epsilon_{fb}^2$ at fixed L . This is the central claim here: the SQL and coherent schemes are not related by a fixed constant-factor speedup (which would give $R(L) = \text{const}$) but by a genuine scaling-class separation — $O(L^2)$ versus $O(L)$ circuit volume — whose ratio diverges as $L \rightarrow \infty$.

$$R(L) \rightarrow \infty \text{ as } L \rightarrow \infty, \quad \text{at any fixed } \epsilon_{fb} > 0. \quad (\text{D19})$$

Step 3 (numeric instantiation at the headline parameters). At $L = 500$, $\epsilon_{fb} = 0.01$, $c = 2$:

$$R(500) = \frac{500}{2 \cdot 2 \cdot (0.01)^2} = \frac{500}{4 \times 10^{-4}} = 1.25 \times 10^6. \quad (\text{D20})$$

This is a ratio of *raw layer-execution counts* — the quantity derived above.

Step 4 (reconciliation with the quoted headline T-count ratio). The headline corroboration quotes a T-count ratio of $\sim 79,000$ between SQL and coherent implementations at the same (L, ϵ_{fb}) . This number is **not** reproduced by (D20) — 1.25×10^6 and 7.9×10^4 differ by roughly a factor of 16, and we do not claim otherwise. The two quantities are different metrics:

- $R(L)$ here counts *layer-executions* (full re-preparation events vs. coherent per-layer control steps) — a circuit-volume proxy built from the two closed-form derivations above.
- The quoted T-count ratio folds in additional, layer-execution-independent factors not modeled here: per-layer gate synthesis cost (rotation synthesis into the Clifford+T gate set), the actual b -dependent cost of the controlled-evolution-plus-uncompute step (noted above as an orthogonal, unmodeled axis), and possibly different constant prefactors in the SQL-side per-shot circuit cost.

Both quantities share the same *scaling class* — linear growth of the SQL/coherent ratio in L (and $1/\epsilon_{fb}^2$ in precision) — which is the structural claim established here. The precise numeric prefactor of 79,000 is a separately computed corroboration (T-count accounting), not re-derived here.

The scaling claim (Eqs. (D18)–(D19)) follows directly from the two preceding results by algebra alone. It does *not* account for the difference between the raw layer-execution ratio 1.25×10^6 and the quoted headline T-count ratio 79,000; that reconciliation requires the per-layer gate-synthesis accounting, which lies outside the scope of the present derivation (SQL total, register width, coherent total). The scaling-class result is what is established here; the exact headline constant is a separate corroboration.

The measure-and-restart $L^2 \rightarrow L$ collapse is FALQON/Hubbard-independent

We show that the mechanism that collapses total circuit volume from $O(L^2)$ to $O(L)$ by replacing measured feedback with coherent feedback depends only on the abstract structure of the feedback loop, not on FALQON’s specific control law or on the Hubbard Hamiltonian; it holds for any measurement-driven iterative algorithm satisfying three named structural conditions.

Step 1 (abstract setup). Let an iterative quantum algorithm run L rounds. Round $k+1$ ’s evolution is parametrised by a classical (scalar, or fixed-size tuple of scalars) quantity θ_{k+1} that is a deterministic function of an estimate of a round- k observable O_k on the state produced by round k :

$$\theta_{k+1} = f(\hat{O}_k), \quad \hat{O}_k \approx \langle O_k \rangle_{\text{state after round } k}. \quad (\text{D21})$$

This is exactly the shape of FALQON’s update ($\theta_{k+1} = \beta_{k+1} = -\langle i[H_d, H_p] \rangle_k$), but nothing in (D21) names H_d , H_p , or the Hubbard lattice: it is satisfied by any control law that is (i) causal (depends only on rounds $\leq k$) and (ii) reduces to a bounded-variance scalar estimation problem per round.

Step 2 (measured branch reproduces the $L^2/2$ count). If $\hat{O}_k$ is obtained by projective measurement, the same two facts used for FALQON apply verbatim and are independent of O_k ’s physical content — shot-noise scaling for any bounded-variance observable, and re-preparation from round 1 because measurement collapses the state:

$$N_{\text{shots}}(\epsilon) \sim 1/\epsilon^2, \quad (\text{D22})$$

$$(\text{executions per shot at round } k) = k. \quad (\text{D23})$$

Eq. (D22) is the central-limit-theorem mean-estimation bound and cites no property of O_k beyond boundedness. Eq. (D23) cites no property beyond causality in (D21):

because θ_{k+1} is not fixed until $\hat{O}_k$ is resolved, and a projective outcome destroys the pre-measurement state, any single shot aimed at round k must re-execute rounds $1, \dots, k$. Summing as in the SQL derivation above gives

$$C_{\text{measured}}(L, \epsilon) = \sum_{k=1}^L \frac{k}{\epsilon^2} = \frac{L(L+1)}{2\epsilon^2} \sim \frac{L^2}{2\epsilon^2}, \quad (\text{D24})$$

identical in form to the FALQON-specific result (as $L \rightarrow \infty$), with O_k and θ_{k+1} left uninterpreted throughout.

Step 3 (the three structural conditions for coherent collapse). The measured branch in Step 2 can be replaced by a single coherent pass if and only if the loop admits three conditions, stated here by name so the scope of the claim is checkable rather than assumed:

C1 *Per-round scalar feedback observable.* O_k is a fixed, round-independent-in-kind observable (or bounded tuple of such) producing a value in a register of bounded width b — the object $\hat{O}_k$ in (D21) is a number, not an unbounded classical program.

C2 *Coherent write.* There exists a unitary W_k that writes an estimate of O_k into an ancilla register without projective measurement, i.e. $W_k(\rho_k \otimes |0\rangle_{\text{anc}})$ correlates the register with O_k 's eigenbasis while leaving ρ_k 's coherences (beyond that correlation) intact.

C3 *Controlled consumption.* The round- $(k+1)$ evolution can be expressed as a register-controlled unitary $U_{k+1}(\text{anc})$ that applies $f(\cdot)$'s effect coherently, consuming the register written by C2 without collapsing it first.

Step 4 (collapse to linear volume under C1–C3). If C1–C3 hold, each round's feedback is resolved by one application of W_k followed by one application of $U_{k+1}(\text{anc})$, both unitary and composed directly onto the running state — no measurement, hence no re-preparation. The circuit executes rounds $1, \dots, L$ exactly once, with per-round control overhead bounded by a constant c (the cost of W_k plus the controlled dispatch, independent of k):

$$C_{\text{coherent}}(L) = \sum_{k=1}^L c = cL = O(L). \quad (\text{D25})$$

Comparing (D24) and (D25): the ratio $C_{\text{measured}}/C_{\text{coherent}} \sim L/(2c\epsilon^2)$ grows without bound in L , so this is a scaling-class separation, not a constant-factor speed-up, and the argument used no fact about H_d , H_p , or the Hubbard lattice at any step.

Scope: this derivation shows that the $L^2 \rightarrow L$ collapse is a consequence of the abstract loop structure (D21) plus C1–C3 — it is a scaling-structure argument. It does *not*

prove that C2 (coherent write) and C3 (controlled consumption) are physically realisable at low cost for an arbitrary O_k : for a generic observable, W_k may require an ancilla register whose width or preparation depth grows with problem size, or $U_{k+1}(\text{anc})$ may not admit an efficient controlled form at all. FALQON's $\langle i[H_d, H_p] \rangle$ satisfies C1–C3 with a 9-bit register and a Trotter-step-scale controlled driver (established in the register-width and coherent-volume derivations above); this shows only that *whenever* C1–C3 hold, the collapse follows, not that they hold universally or cheaply.

Steps 1–4 are exact given C1–C3; the boundary of applicability — whether C2/C3 are cheap for a given O_k — is out of scope and not claimed here.

Specialisation to adaptive (ADAPT-VQE) methods, and the argmax that survives

We show that the class-level $L^2 \rightarrow L$ collapse applies to ADAPT-VQE's outer operator-selection loop, but the per-instance cost is not $c = O(1)$ per round: the feedback is a pool-indexed vector of gradients and its consumption is an argmax, a MAX and not a linear functional. C1–C2 hold cleanly; C3 holds only at an $O(P)$ pool-size prefactor (the coherent comparator/SELECT network), and the inner parameter re-optimisation is a separate loop not removed here.

Step 1 (the ADAPT-VQE loop as a measure-driven iteration). ADAPT-VQE grows an ansatz one operator at a time from a fixed pool of anti-Hermitian generators $\{A_i\}_{i=1}^P$. After macro-iteration k the state is a depth- k product

$$|\psi_k\rangle = \prod_{j=1}^k e^{\theta_j A_{i_j}} |\text{ref}\rangle. \quad (\text{D26})$$

The next operator is chosen by the energy gradient with respect to prepending each pool element. For $A_i^\dagger = -A_i$ and $H^\dagger = H$,

$$g_i^{(k)} = \left. \frac{\partial}{\partial \theta} \right|_{\theta=0} \langle \psi_k | e^{-\theta A_i} H e^{\theta A_i} | \psi_k \rangle = \langle \psi_k | [H, A_i] | \psi_k \rangle, \quad (\text{D27})$$

which is *real*: $[H, A_i]^\dagger = [A_i^\dagger, H^\dagger] = [-A_i, H] = [H, A_i]$, so $[H, A_i]$ is Hermitian. This is the same shape as FALQON's $A_k = \langle i[H_d, H_p] \rangle_k$ (a commutator expectation), evaluated once *per pool element* rather than once per layer. The round- $(k+1)$ unitary is fixed by the selection

$$i^* = \arg \max_{1 \leq i \leq P} |g_i^{(k)}|, \quad \text{append } e^{\theta_{i^*} A_{i^*}}. \quad (\text{D28})$$

Step 2 (measured branch: $P L^2/2$). To read the vector $g^{(k)}$ by projective measurement, each of the P gradients needs $\sim 1/\epsilon^2$ shots (shot-noise mean estimation,

Eq. (D27) is a bounded Hermitian observable), and each shot re-prepares the depth- k state of Eq. (D26) from the reference (measurement collapses $|\psi_k\rangle$). Summing over the L macro-iterations,

$$C_{\text{measured}}(L, \epsilon) = \sum_{k=1}^L \frac{P k}{\epsilon^2} = \frac{P L(L+1)}{2\epsilon^2} \sim \frac{P L^2}{2\epsilon^2}, \quad (\text{D29})$$

the FALQON $L^2/2$ structure of the SQL derivation (as $L \rightarrow \infty$), carrying an explicit pool factor P .

Step 3 (C1–C3 for the operator-selection loop).

C1 Bounded feedback tuple — holds, with a prefactor. The round- k feedback is the P -vector $g^{(k)}$, a bounded tuple of scalars in the sense of C1, held in a register of width Pb (with b the per-gradient precision width from the register-width derivation). It is *not* a single scalar: $P = \text{poly}(N)$ ($O(N) - O(N^4)$ depending on the pool), so the register width is problem-size-dependent, unlike FALQON's fixed 9-bit register.

C2 Coherent write — holds. Each $g_i^{(k)} = \langle [H, A_i] \rangle$ is the expectation of a Hermitian operator (a bounded sum of Pauli strings), written coherently into its b -bit sub-register by the *same* block-encoding / phase-estimation primitive that writes FALQON's $\langle i[H_d, H_p] \rangle$ (coherent-volume derivation above). Doing this for all P operators costs $O(P)$ coherent writes.

C3 Controlled consumption — holds only as a MAX, at $O(P)$. The consumption (D28) is an argmax, which is *not* a linear functional of the register (FALQON's $\beta_{k+1} = -A_k$ is). It is nonetheless realisable coherently as two unitary blocks: a reversible comparator/argmax network $\mathcal{C}$ writing the winning index into an $\lceil \log_2 P \rceil$ -bit index register, $\mathcal{C} : |g^{(k)}\rangle|0\rangle_{\text{idx}} \mapsto |g^{(k)}\rangle|i^*\rangle$, followed by a SELECT multiplexer over the pool,

$$\text{SEL}(\theta) = \sum_{i=1}^P |i\rangle\langle i|_{\text{idx}} \otimes e^{\theta A_i}, \quad (\text{D30})$$

which applies the chosen exponential controlled on the index register. Both are unitary, so no measurement and no re-preparation — but $\mathcal{C}$ costs $O(Pb)$ reversible-comparator gates and ancillas and SEL costs $O(P)$ controlled exponentials.

Step 4 (collapse in L , with the pool prefactor named). Under C1–C3 each macro-iteration is one coherent pass: coherent-write $g^{(k)}$, run $\mathcal{C}$, apply $\text{SEL}(\theta_{i^*})$, uncompute. There is no collapse, so rounds $1, \dots, L$ execute once:

$$C_{\text{coherent}}(L) = \sum_{k=1}^L c_P = c_P L = O(PL), \quad c_P = O(Pb), \quad (\text{D31})$$

and the separation is

$$\frac{C_{\text{measured}}}{C_{\text{coherent}}} \sim \frac{P L^2 / (2\epsilon^2)}{c_P L} = \frac{L}{2c'\epsilon^2}, \quad c' = c_P / P = O(b), \quad (\text{D32})$$

which still *grows linearly in L* — the $L^2 \rightarrow L$ collapse in the ansatz depth L is real and instance-independent. The pool size P cancels from the *ratio* (D32) (it multiplies both branches), but it does not vanish from the *coherent cost* (D31): the per-round constant is $c_P = O(Pb)$, not the $O(1)$ of FALQON.

Limitations.

- *The argmax is a MAX, not a linear functional.* FALQON consumes its register by a single register-controlled *rotation* $e^{-i\beta H_d}$ (linear in the register). ADAPT-VQE consumes its register by *selection* (D28). Coherent selection survives as the comparator/SELECT network of Step 3, so classical selection is *displaced into a coherent $O(P)$ sub-circuit*, not eliminated: this is exactly where classical selection “survives” inside the coherent loop.
- *Pool-size prefactor.* The coherent per-round cost carries $c_P = O(Pb)$, $P = \text{poly}(N)$; the collapse is a scaling-class win in L with a problem-size prefactor, not the flat $O(1)$ -per-round of the FALQON instance.
- *Inner re-optimisation not removed.* After each append ADAPT-VQE re-optimises *all* θ_j by an inner VQE — a separate measure-driven optimisation loop. The step angle θ_{i^*} is register-controlled (hence linear, closable) only if fixed by a single gradient/line-search step; full inner re-optimisation is outside the present scope, which removes the *outer* operator-selection restart alone.

In summary, C1–C2 are exact; C3 is realised as a coherent argmax at an explicit $O(P)$ pool prefactor. The $L^2 \rightarrow L$ collapse of the outer selection loop holds instance-independently, subject to the surviving classical-selection cost, the pool prefactor, and the un-removed inner re-optimisation.

Scope of the collapse for this case. Unlike FALQON and the ADAPT-VQE outer loop, iterative phase estimation carries no cumulative re-preparation cost: because the target is a U -eigenstate, measuring the round- k ancilla does not collapse the system state and the lower bits are classical feedforward. The measured cost for iterative phase estimation is therefore *already* $O(L)$; the quadratic overhead removed by the $L^2 \rightarrow L$ reduction arises only for feedback taken on non-eigenstate intermediates, and the Griffiths–Niu semiclassical-QFT identity relates the measured and coherent branches as $O(L) = O(L)$. What holds unconditionally here is the primitive identification (C1–C3 below): the feedback angle is linear in the bit register and the classically-controlled rotation cascade is

exactly the coherent inverse-QFT primitive. IPE thus realises the coherent-feedback primitive at no collapse cost, precisely because its feedback observable never lives on a collapsing intermediate.

Iterative phase estimation: the Kitaev bit-controlled rotation is the coherent-feedback primitive

We show that for iterative phase estimation (IPE) the class conditions C1–C3 hold in their strongest form — the per-round feedback is a single measured bit, and the rotation it conditions is a linear functional of the accumulated bit register — so the measured (semiclassical) loop and the coherent register-controlled loop are the two sides of the semiclassical-QFT equivalence, with no argmax and no pool prefactor. Because the target is a U -eigenstate the measured cost is already $O(L)$; the $L^2/2 \rightarrow L$ reduction applies to feedback taken on non-eigenstate intermediates. The only additional cost is a space–time trade (one recycled ancilla $\rightarrow L$ simultaneous ancillas) and the standard eigenstate/readout assumption.

Step 1 (the sequential-bit loop). IPE estimates the phase $\varphi \in [0, 1)$ of $U|\psi\rangle = e^{2\pi i\varphi}|\psi\rangle$, $\varphi = 0.\varphi_1\varphi_2\cdots\varphi_L$ in binary, resolving one bit per round from least significant up. Round k places an ancilla in $|+\rangle$, applies the controlled power $\Lambda(U^{2^{L-k}})$ (phase kickback), applies a feedback rotation $R_z(\omega_k)$, then H and a measurement giving bit b_k . The feedback angle is fixed by all previously measured bits:

$$\omega_k = -2\pi (0.0b_{k-1}b_{k-2}\cdots b_1) = -2\pi \sum_{\ell=1}^{k-1} b_\ell 2^{-(k-\ell+1)}. \quad (\text{D33})$$

This is precisely the shape of the class setup $\theta_{k+1} = f(\hat{O}_k)$: the next unitary depends on an observable (here a bit) of the current state.

Step 2 (measured branch: the general $L^2/2$ comparator). For feedback taken on a non-eigenstate intermediate, measurement-and-restart would resolve each bit to confidence ϵ_{fb} by $\sim 1/\epsilon_{\text{fb}}^2$ repetitions, each re-preparing the round's input by re-running the earlier phase-kickback rounds since the projective readout destroyed the coherent ancilla chain, giving round- k cost k/ϵ_{fb}^2 and

$$C_{\text{measured}}(L, \epsilon_{\text{fb}}) = \sum_{k=1}^L \frac{k}{\epsilon_{\text{fb}}^2} = \frac{L(L+1)}{2\epsilon_{\text{fb}}^2} \sim \frac{L^2}{2\epsilon_{\text{fb}}^2}, \quad (\text{D34})$$

the same $L^2/2$ as the SQL derivation, with no pool factor: the feedback is one bit, not a P -vector. For IPE's eigenstate target this re-preparation is avoided — the lower bits are classical feedforward and the eigenstate is not collapsed — so the measured cost is already $O(L)$, and (D34) serves only as the general-case comparator.

Step 3 (C1–C3 in their strongest form).

C1 Per-round scalar — the strongest case. The feedback $b_k \in \{0, 1\}$ is a *single bit*; the register is the accumulated $|b_1 \cdots b_L\rangle$, width $L = \lceil \log_2(1/\epsilon_\varphi) \rceil$. This is C1 at its floor: not a tuple, a bit.

C2 Coherent write — trivial. No separate write oracle is needed: the round- k ancilla, if left *unmeasured*, already *is* the register qubit holding b_k coherently after its Hadamard. C2 is discharged by simply not measuring.

C3 Controlled consumption — a LINEAR functional, exactly the primitive. The angle (D33) is linear in the bits, so the classically-controlled $R_z(\omega_k)$ becomes a product of *quantum*-controlled rotations, one per prior ancilla:

$$R_z(\omega_k)|_{b_\ell \text{ classical}} \mapsto \prod_{\ell=1}^{k-1} \Lambda_\ell \left(R_z(-2\pi 2^{-(k-\ell+1)}) \right), \quad (\text{D35})$$

each controlled on ancilla ℓ . Eq. (D35) is register-controlled evolution with the angle a linear functional of the register — FALQON's $e^{-i\beta H_d}$ primitive in its purest form. The full cascade over all rounds is exactly the coherent inverse quantum Fourier transform.

Step 4 (collapse, and the semiclassical-QFT equivalence). Keeping every ancilla coherent and replacing (D33)'s measured feedback by the controlled cascade (D35), the algorithm runs the L phase-kickback rounds once, applies the coherent inverse QFT, and reads out — a single coherent pass:

$$C_{\text{coherent}}(L) = \sum_{k=1}^L c = cL = O(L), \quad (\text{D36})$$

$$\frac{C_{\text{measured}}}{C_{\text{coherent}}} \sim \frac{L}{2c\epsilon_{\text{fb}}^2} \xrightarrow{L \rightarrow \infty} \infty.$$

That the measured cascade (D33) and the coherent cascade (D35) compute the same transform is the Griffiths–Niu semiclassical-QFT theorem. The ratio in (D36) is the general-case comparator (measured cost $\sim L^2$ for non-eigenstate feedback); for IPE's eigenstate target the measured cost is already $O(L)$, so the two branches are the $O(L) = O(L)$ semiclassical-QFT equivalence rather than a scaling separation. The per-round constant c is genuinely $O(1)$ (one controlled rotation of fixed angle per prior bit; the cascade is $O(L)$ gates total, with no hidden pool factor), and there is no argmax and no P .

Limitation.

- *Space–time trade, not a free lunch.* The measured loop recycles *one* ancilla across rounds; the coherent loop holds $L = \log_2(1/\epsilon_\varphi)$ ancillas simultaneously (the full register). The $L^2 \rightarrow L$ time collapse buys itself $O(L)$ coherent qubits of width. The register width $L = \log(1/\epsilon_\varphi)$ is modest, but the trade is real and is the boundary of this case.

- *Eigenstate and final readout.* IPE assumes $|\psi\rangle$ is (close to) a U -eigenstate; for a superposition the coherent register entangles with the eigenphases and the phase is read by *one* terminal measurement of the register — consistent with the thesis (measurement confined to the single load-bearing readout), but it is not zero measurement. Robustness of the leading analog bit (majority vote in practice) is folded into that terminal readout.

C1–C3 hold in their strongest form, and the equivalence of the measured and coherent branches is the Griffiths–Niu semiclassical-QFT theorem, exact, with $c = O(1)$ and no pool prefactor. The space–time trade and the eigenstate/terminal-readout assumption are the boundary of this case.

Error-adaptive compilation: the two-ordering heralding-bit witness, generalised to a per-checkpoint family

We show that error-adaptive compilation (choosing the next segment’s compilation from a run-time error observable of the current segment) has, as its coherent instance, the two-ordering Trotter-error witness of Sec. VIII. Its feedback is a single heralding bit written coherently by the LCU interferometer (C2) and consumed by a flag-controlled correction (C3, a 1-bit-controlled unitary). The quiet branch gives a per-checkpoint cost of $O(1)$ with no restart; the loud branch is the limiting case, its corrective cost left unspecified.

Step 1 (error-adaptive compilation as a measure-driven iteration). Partition a Trotterised evolution into M checkpoints (segments). At checkpoint k the compiler reads an error indicator of the segment just executed and chooses the next segment’s compilation (keep dt , halve it, or substitute a higher-order step). Writing the two orderings of one Trotter step as U_{AB} (layer order A then B) and U_{BA} (reversed), the error indicator is the flag probability of this paper’s witness,

$$\begin{aligned} P(\text{flag}) &= \frac{1}{2} \left(1 - \text{Re} \langle U_{BA}^\dagger U_{AB} \rangle \right) \\ &= \frac{dt^4}{4} \langle K^\dagger K \rangle + O(dt^5), \quad K = [A, B], \end{aligned} \quad (\text{D37})$$

(both derived below). The next compilation depends on this observable of the current state: the class shape $\theta_{k+1} = f(\hat{O}_k)$ with $\hat{O}_k$ the heralding bit.

Step 2 (measured branch: the $M^2/2$ structure). Estimating the per-checkpoint error by projective measurement to confidence ϵ_{fb} needs $\sim 1/\epsilon_{\text{fb}}^2$ shots, each re-preparing the state at checkpoint k by re-running segments $1, \dots, k$ (the readout collapsed it). Summing over checkpoints reproduces the SQL derivation with M in

place of L :

$$C_{\text{measured}}(M, \epsilon_{\text{fb}}) = \sum_{k=1}^M \frac{k}{\epsilon_{\text{fb}}^2} = \frac{M(M+1)}{2\epsilon_{\text{fb}}^2} \sim \frac{M^2}{2\epsilon_{\text{fb}}^2}. \quad (\text{D38})$$

Step 3 (C1–C3 — the witness is the coherent write).

C1 Per-checkpoint scalar — a single bit. The feedback is the flag $f_k \in \{0, 1\}$ of Eq. (D37), one bit per checkpoint (graded refinement would widen this; see the gaps). C1 holds at its floor, as for IPE.

C2 Coherent write — already constructed. The two-ordering LCU interferometer of the main text writes $P(\text{flag})$ into the ancilla *without* measuring it: the flag ancilla holds the error indicator coherently. C2 is not a new requirement for this case — it is the same witness, run with its Hadamard-and-measure deferred.

C3 Controlled consumption — a 1-bit-controlled correction. With the flag held coherently, the refinement is a flag-controlled unitary,

$$V_k = |0\rangle\langle 0|_{\text{flag}} \otimes \mathbb{I} + |1\rangle\langle 1|_{\text{flag}} \otimes \mathcal{R}_k, \quad (\text{D39})$$

$\mathcal{R}_k$ the corrective step (a finer sub-step, or the quiet-branch higher-order substitution). This is register-controlled evolution controlled on a single bit — the minimal, most linear case of C3, no argmax. In the *quiet* branch ($f_k = 0$) the symmetrised step $(U_{AB} + U_{BA})/2$ has already advanced the state to $O(dt^3)$ accuracy (derived below), so the identity arm of (D39) is correct *by construction*: detection and leading-error cancellation happen in the same coherent step, and no correction is applied.

Step 4 (per-checkpoint cost at the family level). Running the witness in-line, checkpoint k costs, over its host segment, exactly the per-checkpoint witness cost (derived below) lifted to this case: one ancilla (already carried by the coherent LCU register), $\sim$ one Trotter-step depth for the interferometer, one flag-controlled correction (D39), and — crucially — *no re-preparation*, because nothing is measured. Per checkpoint this is

$$\begin{aligned} c_{\text{chk}} &= \underbrace{1}_{\text{ancilla (shared)}} + \underbrace{O(1) \text{ Trotter-step depth}}_{\text{interferometer}} \\ &\quad + \underbrace{[f_k = 1] \|\mathcal{R}_k\|}_{\text{correction, loud only}} = O(1) \text{ (quiet branch)}, \end{aligned} \quad (\text{D40})$$

so the family cost is the single coherent pass

$$\begin{aligned} C_{\text{coherent}}(M) &= \sum_{k=1}^M c_{\text{chk}} = O(M), \\ \frac{C_{\text{measured}}}{C_{\text{coherent}}} &\sim \frac{M}{2\epsilon_{\text{fb}}^2} \xrightarrow{M \rightarrow \infty} \infty. \end{aligned} \quad (\text{D41})$$

The witness rides *inside* the coherent loop at one heralding bit per checkpoint, generalising the FALQON-specific witness-cost result (below) to any Trotterised measure-driven algorithm: the $M^2 \rightarrow M$ collapse holds for the whole error-adaptive class, amortised across the m steps per checkpoint as $c'M$ with $c' < c$.

Limitations.

- *The loud branch is not $O(1)$, and no correction protocol is specified.* Eq. (D40)'s $O(1)$ per-checkpoint cost is the *quiet-branch* claim. When the flag fires, $\mathcal{R}_k$ (finer dt , redo, higher-order patch) can cost more than one step, and the main text is explicit that a fired flag is “a detection probability, not a correction ... no correction protocol is specified here.” The loud-branch corrective cost is the same limitation identified for the witness-cost derivation below, now at the family level: what is established is the *detection-and-ride* cost, not the correction cost.
- *Threshold vs. graded refinement.* A binary herald consumes the register by the 1-bit-controlled (D39) — no argmax. But adaptively choosing a *continuous* dt from a graded error estimate reintroduces a wider register and a SELECT over correction strengths, inheriting the $O(P)$ argmax cost of the adaptive-VQE case above. The $O(1)$ claim is specific to the binary herald.
- *Acceptance-probability dependence.* The amortised $c'M$ assumes the quiet branch dominates (high cumulative acceptance, e.g. 0.9996 at $dt = 0.05$); at coarse dt the flag fires often (0.341 at $dt = 0.8$) and the loud-branch cost is no longer negligible. The family collapse is a claim about the *quiet-dominated* regime.

C1–C3 are realised using the two-ordering witness as the coherent write; the per-checkpoint $O(1)$ detect-and-ride cost and the $M^2 \rightarrow M$ family collapse hold for the quiet branch. The loud-branch correction cost, graded-refinement selection, and acceptance-probability dependence are the limitations of this case.

Flag-branch probability of the controlled-ordering witness

We show that $P(\text{flag}) = \frac{1}{2}(1 - \text{Re}\langle U_{BA}^\dagger U_{AB} \rangle)$ for the two-ancilla controlled-ordering interferometer, and that this probability is exactly zero when $[A, B] = 0$.

a Setup. Let $U_{AB} = e^{-iAdt}e^{-iBdt}$ and $U_{BA} = e^{-iBdt}e^{-iAdt}$ act on a system register prepared in $|\psi\rangle$. The witness circuit realises, on a two-ancilla register, a one-hot encoding of a single logical control qubit: the physical state $|10\rangle_{\text{anc}}$ (ancilla 0 excited) plays the role of logical $|1\rangle_c$, gating U_{AB} , and $|01\rangle_{\text{anc}}$ (ancilla 1 excited) plays the role of logical $|0\rangle_c$, gating U_{BA} ; the

closing Hadamards act identically on this logical qubit. Under this correspondence the circuit is the standard Hadamard-test comparator for two unitaries, which we now analyse on the logical control qubit c .

b Step 1 — opening interference. Starting from $|0\rangle_c|\psi\rangle$, the opening Hadamard H_c produces an equal-weight superposition of the two order-branches:

$$H_c|0\rangle_c|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle_c + |1\rangle_c)|\psi\rangle. \quad (\text{D42})$$

c Step 2 — controlled orderings. Applying U_{BA} controlled on $c = 0$ and U_{AB} controlled on $c = 1$ gives

$$\frac{1}{\sqrt{2}}(|0\rangle_c U_{BA}|\psi\rangle + |1\rangle_c U_{AB}|\psi\rangle). \quad (\text{D43})$$

d Step 3 — closing Hadamard. Using $H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, $H|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$, the state becomes

$$\frac{1}{2}(|0\rangle_c(U_{BA} + U_{AB})|\psi\rangle + |1\rangle_c(U_{BA} - U_{AB})|\psi\rangle). \quad (\text{D44})$$

The $c = 1$ (“flag”) outcome carries the *difference* of the two orderings and is exactly the destructive port of the interferometer; the $c = 0$ (“quiet”) outcome carries the sum.

e Step 4 — Born rule on the flag outcome. By definition,

$$\begin{aligned} P(\text{flag}) &= \frac{1}{4} \|(U_{BA} - U_{AB})|\psi\rangle\|^2 \\ &= \frac{1}{4} \left[\langle \psi | U_{BA}^\dagger U_{BA} | \psi \rangle + \langle \psi | U_{AB}^\dagger U_{AB} | \psi \rangle \right. \\ &\quad \left. - \langle \psi | U_{BA}^\dagger U_{AB} | \psi \rangle - \langle \psi | U_{AB}^\dagger U_{BA} | \psi \rangle \right]. \end{aligned} \quad (\text{D45})$$

Since U_{AB} and U_{BA} are each unitary, the first two terms are each 1. For the cross terms, unitarity of the inner product gives $\langle \psi | U_{AB}^\dagger U_{BA} | \psi \rangle = (\langle \psi | U_{BA}^\dagger U_{AB} | \psi \rangle)^*$, so their sum is $2 \text{Re}\langle U_{BA}^\dagger U_{AB} \rangle$, where $\langle X \rangle \equiv \langle \psi | X | \psi \rangle$. Hence

$$\begin{aligned} P(\text{flag}) &= \frac{1}{4} [2 - 2 \text{Re}\langle U_{BA}^\dagger U_{AB} \rangle] \\ &= \frac{1}{2} (1 - \text{Re}\langle U_{BA}^\dagger U_{AB} \rangle). \end{aligned} \quad (\text{D46})$$

This is the stated result.

f Step 5 — exact darkness when $[A, B] = 0$. If $[A, B] = 0$ then e^{-iAdt} and e^{-iBdt} commute for every dt (two operators generated by commuting Hermitian generators commute), so

$$\begin{aligned} [A, B] = 0 &\implies U_{AB} = e^{-iAdt}e^{-iBdt} \\ &= e^{-iBdt}e^{-iAdt} = U_{BA}. \end{aligned} \quad (\text{D47})$$

Then $U_{BA}^\dagger U_{AB} = U_{AB}^\dagger U_{AB} = \mathbb{I}$, so $\langle U_{BA}^\dagger U_{AB} \rangle = \langle \psi | \psi \rangle = 1$ and $\text{Re}\langle U_{BA}^\dagger U_{AB} \rangle = 1$. Substituting into Eq. (D46),

$$P(\text{flag}) = \frac{1}{2}(1 - 1) = 0 \quad \text{whenever } [A, B] = 0. \quad (\text{D48})$$

This step closes fully: it is state-independent (holds for every $|\psi\rangle$), hence at the operator level).

g Remark. The only modelling input is the two-ancilla one-hot $\leftrightarrow$ single-logical-control-qubit identification used to map the physical circuit onto the standard Hadamard-test comparator; the Born-rule algebra downstream of that identification is exact.

Leading-order dt^4 scaling of the flag-branch probability

We show that to leading order in dt , $P(\text{flag}) \sim \frac{dt^4}{4} \langle K^\dagger K \rangle$ with $K \equiv [A, B]$, so that $P(\text{flag})$ is quartic in the step size, matching the empirically fitted log-log slope 3.949 against the theoretical exponent 4.

h Step 1 — BCH expansion of each ordering. Writing $X = -iAdt$, $Y = -iBdt$ and using $\ln(e^X e^Y) = X + Y + \frac{1}{2}[X, Y] + O(dt^3)$ with $[X, Y] = -dt^2[A, B]$,

$$\begin{aligned} \ln U_{AB} &= -i(A+B)dt - \frac{dt^2}{2}[A, B] + O(dt^3), \\ \ln U_{BA} &= -i(A+B)dt + \frac{dt^2}{2}[A, B] + O(dt^3), \end{aligned} \quad (\text{D49})$$

the sign flip on the dt^2 term coming from $[Y, X] = -[X, Y]$. Both generators are exactly anti-Hermitian (nested commutators of anti-Hermitian generators stay anti-Hermitian), consistent with U_{AB}, U_{BA} unitary.

i Step 2 — the relative generator. Since U_{BA} is unitary, $U_{BA}^\dagger = \exp(-\ln U_{BA})$. Composing via BCH once more, the $O(dt)$ pieces cancel between $-\ln U_{BA}$ and $\ln U_{AB}$, and the $[-\ln U_{BA}, \ln U_{AB}]$ cross-commutator is itself $O(dt^3)$ (each entry is $O(dt)$ at leading order and they are proportional to the same generator $A+B$ at that order, so their leading commutator vanishes). What survives at second order is

$$U_{BA}^\dagger U_{AB} = \exp(-dt^2[A, B] + O(dt^3)). \quad (\text{D50})$$

(Sign/normalization remark: this is the same statement as the schematic $\exp(i dt^2[A, B] + \dots)$ once the anti-Hermitian generator $[A, B]$ is repackaged as $i dt^2(i[A, B])$; we work directly with $K \equiv [A, B]$ below since it makes the positivity of the leading term manifest without an extra factor of i .)

j Step 3 — expand the exponential and take the expectation. Let $K \equiv [A, B]$ (anti-Hermitian: $K^\dagger = -K$) and write the exponent of Eq. (D50) as $M = -dt^2 K + R$ with $R = O(dt^3)$ anti-Hermitian. Expanding,

$$U_{BA}^\dagger U_{AB} = \mathbb{I} - dt^2 K + R + \frac{dt^4}{2} K^2 + O(dt^5), \quad (\text{D51})$$

using $M^2 = dt^4 K^2 + O(dt^5)$ (cross terms KR are $O(dt^5)$). Taking expectation value in $|\psi\rangle$,

$$\langle U_{BA}^\dagger U_{AB} \rangle = 1 - dt^2 \langle K \rangle + \langle R \rangle + \frac{dt^4}{2} \langle K^2 \rangle + O(dt^5). \quad (\text{D52})$$

k Step 4 — the real part kills the odd-order (imaginary) terms. Because K is anti-Hermitian, $\langle K \rangle$ is purely imaginary for any $|\psi\rangle$ (if $X^\dagger = -X$ then $\langle X \rangle^* = \langle X^\dagger \rangle = -\langle X \rangle$). The same holds for the $O(dt^3)$ term $\langle R \rangle$, since R is itself anti-Hermitian by Step 1. Hence

$$\text{Re}(-dt^2 \langle K \rangle) = 0, \quad \text{Re} \langle R \rangle = 0. \quad (\text{D53})$$

For the surviving dt^4 term, $K^\dagger K = -K^2$ is Hermitian and positive semi-definite (it is $K^\dagger K$), so $\langle K^2 \rangle = -\langle K^\dagger K \rangle$ is real and non-positive. Thus

$$\text{Re} \langle U_{BA}^\dagger U_{AB} \rangle = 1 - \frac{dt^4}{2} \langle K^\dagger K \rangle + O(dt^5). \quad (\text{D54})$$

l Step 5 — assemble $P(\text{flag})$. Substituting Eq. (D54) into the flag-branch result $P(\text{flag}) = \frac{1}{2}(1 - \text{Re} \langle U_{BA}^\dagger U_{AB} \rangle)$ (derived above),

$$\begin{aligned} P(\text{flag}) &= \frac{1}{2} \left(\frac{dt^4}{2} \langle K^\dagger K \rangle + O(dt^5) \right) \\ &= \frac{dt^4}{4} \langle K^\dagger K \rangle + O(dt^5), \quad K = [A, B]. \end{aligned} \quad (\text{D55})$$

Since $K^\dagger K \succeq 0$, this leading term is manifestly non-negative, as a probability must be. Equation (D55) is the quartic scaling.

m Empirical corroboration. The fitted log-log slope of $P(\text{flag})$ vs. dt is reported at 3.949, against the theoretical exponent 4 derived above; the $\sim 1.3\%$ deviation is consistent with the neglected $O(dt^5)$ correction and finite- dt fit range, not with a different leading power.

n Remark. The single normalization choice ($K = [A, B]$ rather than iK) is stated in Step 2 and does not affect the magnitude $\langle K^\dagger K \rangle$.

Quiet-branch symmetrisation raises accuracy from $O(dt^2)$ to $O(dt^3)$

We show that the bare (single-ordering) first-order Trotter step U_{AB} has local error $O(dt^2)$ against the exact propagator, while the symmetrised quiet-branch combination $(U_{AB} + U_{BA})/2$ has local error $O(dt^3)$, because the leading $dt^2[A, B]$ error term enters U_{AB} and U_{BA} with opposite signs and cancels.

o Step 1 — Taylor-expand the bare step. To $O(dt^3)$,

$$\begin{aligned} e^{-iAdt} &= \mathbb{I} - iAdt - \frac{A^2 dt^2}{2} + O(dt^3), \\ e^{-iBdt} &= \mathbb{I} - iBdt - \frac{B^2 dt^2}{2} + O(dt^3). \end{aligned} \quad (\text{D56})$$

Multiplying,

$$\begin{aligned} U_{AB} &= e^{-iAdt} e^{-iBdt} \\ &= \mathbb{I} - i(A+B)dt \\ &\quad - \frac{dt^2}{2} (A^2 + B^2 + 2AB) + O(dt^3). \end{aligned} \quad (\text{D57})$$

p Step 2 — Taylor-expand the exact propagator. Similarly,

$$\begin{aligned} e^{-i(A+B)dt} &= \mathbb{I} - i(A+B)dt - \frac{dt^2}{2}(A+B)^2 + O(dt^3) \\ &= \mathbb{I} - i(A+B)dt \\ &\quad - \frac{dt^2}{2}(A^2 + B^2 + AB + BA) + O(dt^3). \end{aligned} \quad (\text{D58})$$

q Step 3 — bare local error is $O(dt^2)$. Subtracting Eq. (D58) from Eq. (D57), the $\mathbb{I}$ and $O(dt)$ terms cancel exactly, and $A^2 + B^2$ cancels from the $O(dt^2)$ piece, leaving

$$\begin{aligned} U_{AB} - e^{-i(A+B)dt} &= -\frac{dt^2}{2}(2AB - AB - BA) + O(dt^3) \\ &= -\frac{dt^2}{2}[A, B] + O(dt^3). \end{aligned} \quad (\text{D59})$$

Since $[A, B] \neq 0$ generically, Eq. (D59) is a genuine, non-vanishing $O(dt^2)$ term: the bare first-order Trotter step has local error $O(dt^2)$.

r Step 4 — the reversed ordering carries the opposite sign. Repeating Steps 1–3 with $A \leftrightarrow B$ swapped (i.e. for $U_{BA} = e^{-iBdt}e^{-iAdt}$ against the same exact propagator, which is symmetric under $A \leftrightarrow B$) gives, by the antisymmetry $[B, A] = -[A, B]$,

$$\begin{aligned} U_{BA} - e^{-i(A+B)dt} &= -\frac{dt^2}{2}[B, A] + O(dt^3) \\ &= +\frac{dt^2}{2}[A, B] + O(dt^3). \end{aligned} \quad (\text{D60})$$

The two orderings' leading error terms are exactly equal in magnitude and opposite in sign.

s Step 5 — the symmetrised (quiet-branch) combination. Averaging Eqs. (D59) and (D60),

$$\begin{aligned} \frac{U_{AB} + U_{BA}}{2} - e^{-i(A+B)dt} &= \frac{1}{2} \left[-\frac{dt^2}{2}[A, B] + \frac{dt^2}{2}[A, B] \right] + O(dt^3) \\ &= O(dt^3). \end{aligned} \quad (\text{D61})$$

The dt^2 terms cancel identically (not approximately — they are the same operator $\frac{dt^2}{2}[A, B]$ with opposite signs), so the symmetrised step matches the exact propagator through $O(dt^2)$ and its local error is $O(dt^3)$: one order higher than either bare ordering alone. This is the single-step analogue of the Strang/symmetric-splitting (multi-product) accuracy gain.

t Step 6 — relation to the quiet-branch amplitude. From the flag-probability derivation above, the quiet ($c = 0$) output amplitude of the interferometer is $\frac{1}{2}(U_{BA} +$

$U_{AB})|\psi\rangle$, i.e. exactly the symmetrised combination analysed above (up to the overall normalisation $\frac{1}{2}$ carried by the closing Hadamard, which does not affect the order of the error). Its deviation from the ideal evolved state $e^{-i(A+B)dt}|\psi\rangle$ is therefore $O(dt^3)$ by Eq. (D61), one full order better than the bare step's $O(dt^2)$.

u Scope. This establishes the *operator-error* orders $O(dt^2)$ (bare) and $O(dt^3)$ (symmetrised); it does not itself derive the reported *infidelity* exponents. Infidelity is quadratic in the leading error operator to leading order ($1 - |\langle\psi_{\text{exact}}|\psi_{\text{approx}}\rangle|^2 \sim \|\delta\|^2$ for small error operator δ), which is consistent with, but not a full derivation of, the reported empirical infidelity slopes 3.951 (bare, $\approx 2 \times 2$) and 5.907 (quiet, $\approx 2 \times 3$) against the amplitude-level exponents 2 and 3 derived here. Promoting this consistency check to a derivation would require controlling the infidelity's leading-order coefficient (normalisation of δ , overlap with $|\psi\rangle$), which is left open here.

v Remark. Steps 1–6 establish the operator-error scaling. One limitation remains: the amplitude-to-infidelity exponent doubling is noted as consistent but not separately derived here.

The two-ordering witness rides inside the coherent feedback loop at no leading-order cost

We show that the per-checkpoint cost of the two-ordering Trotter-error witness is a bounded $O(1)$ additive quantity (one ancilla already carried by the coherent construction, one Trotter-step of rotation depth, one heralding readout) that, amortised over the layers between checkpoints and combined with the no-restart quiet branch, leaves the host loop's $O(L)$ scaling class untouched.

Step 1 (per-checkpoint cost). Inserted between two chosen FALQON layers, the two-ordering witness (the U_{AB}/U_{BA} interferometer, derived above) costs, under the surface-code / Clifford+T accounting used throughout:

$$(\text{ancilla}) = 1, (\text{rotation depth}) \approx d_{\text{layer}}, (\text{readout}) = 1, \quad (\text{D62})$$

where $d_{\text{layer}} \approx 339$ rotations is the depth of one Trotter step — the same rotation count as one FALQON layer (339 R_z rotations $\times$ 100 T-gates/rotation = 33,900 T-gates/layer, so the witness's added depth is ≈ 1 layer's worth of rotations, i.e. $O(1)$ in the layer count L).

Step 2 (the ancilla is already paid for). The coherent feedback construction (derived above) already carries a $b = 9$ -bit ancilla register through the whole run, written coherently each layer and consumed by the register-controlled driver. The witness's single ancilla in (D62) is drawn from — or added to — this pre-existing coherent register rather than requiring a new one to be prepared, threaded, and later discarded from scratch:

$$\Delta(\text{ancilla count}) = 0 \quad (\text{D63})$$

marginally, relative to the coherent LCU construction already in place. This is a bookkeeping fact, not a new physical mechanism: it holds because both the feedback register and the witness flag are ancillary qubits managed by the same coherent-control machinery, so the witness introduces no ancilla the run was not already resourced for.

Step 3 (amortised fractional depth overhead). Let checkpoints be placed every m layers ($m \geq 1$). Between two checkpoints the host loop executes m layers at depth d_{layer} each, i.e. $m d_{\text{layer}}$, to which one checkpoint adds d_{layer} :

$$\text{overhead fraction} = \frac{d_{\text{layer}}}{m \cdot d_{\text{layer}}} = \frac{1}{m}. \quad (\text{D64})$$

Over the full L -layer run there are L/m checkpoints, adding total depth $(L/m) d_{\text{layer}}$ against a host depth of $L d_{\text{layer}}$, giving the same ratio $1/m$ independent of L — the overhead is a per-checkpoint constant, amortised, not a term that grows with the run length.

Step 4 (no restart on the quiet outcome preserves the host's linear scaling). The witness readout in (D62) is a measurement of the single heralding ancilla only — not of the feedback register that drives round-to-round control (the coherent-volume derivation's $C3$ consumption remains coherent throughout). On the quiet outcome the run proceeds on the symmetrised, error-improved step (derived below) with no re-preparation: a quiet outcome triggers no restart, so the coherence of the host loop's state is preserved across the checkpoint. Because the checkpoint neither collapses the feedback register (Step 2) nor forces re-execution from layer 1 (Step 4), the host loop's per-layer cost structure established for the coherent branch — $C_{\text{coherent}}(L) = cL$, the linear result derived above — is unmodified by the insertion of checkpoints; only the constant c is shifted:

$$\begin{aligned} C_{\text{coherent+witness}}(L) &= cL + \frac{L}{m} d_{\text{layer}} \\ &= \left(c + \frac{d_{\text{layer}}}{m} \right) L = c' L = O(L). \end{aligned} \quad (\text{D65})$$

Step 5 (conclusion). Eqs. (D62)–(D65) show the witness's cost is $O(1)$ per checkpoint (one already-carried ancilla, one layer-scale rotation depth, one bit read out), amortises to a fractional overhead $1/m$ that does not grow with L , and — because the quiet branch requires no restart — leaves the exponent of the host loop's scaling law unchanged: $O(L) \rightarrow O(L)$, with only the leading constant modified from c to $c' = c + d_{\text{layer}}/m$.

Scope. This accounts for the checkpoint's ancilla, depth, and readout cost and the no-restart behaviour *on the quiet outcome*. It does not derive how often the quiet outcome occurs (that is $P(\text{flag}) \sim dt^4 \langle K^\dagger K \rangle / 4$, derived below separately) nor specify the corrective procedure on

a loud (flagged) outcome; the flagged branch is outside the present scope and is not asserted to be free.

The additive $O(1)$ -per-checkpoint result and the preserved $O(L)$ scaling class follow from (D62)–(D65) for the quiet branch; the loud-outcome cost is not covered here.

Numerical reproduction of the SQL/AE/coherent resource-gap ratios

We recompute the headline resource-gap ratios at $L = 500$, $\epsilon_{fb} = 0.01$, and the claimed L - and ϵ_{fb} -scaling laws, independently from the reported T -counts and wall-clock figures; they reproduce to within two significant figures.

Ratios. From $T_{\text{SQL}} = 4.2375 \times 10^{14}$, $T_{\text{AE}} = 6.656 \times 10^{11}$, $T_{\text{COHERENT}} = 5.342 \times 10^9$ and wall clocks 2502.1 yr / 187.0 h (converted at 1 yr = 8766 h), recomputation gives $T_{\text{SQL}}/T_{\text{AE}} = 637$ (≈ 640), $T_{\text{AE}}/T_{\text{COHERENT}} = 125$ (≈ 120), $T_{\text{SQL}}/T_{\text{COHERENT}} = 79,300$ ($\approx 79,000$), and wall-clock $\text{SQL}/\text{COHERENT} = 117,300$ ($\approx 120,000$); all four match the manuscript's stated values at 2 s.f. (max relative error 3.8%).

Scaling. COHERENT wall clock vs. L (at $\epsilon_{fb} = 0.01$) fits a power law of exponent ≈ 1.06 across both $100 \rightarrow 500$ and $500 \rightarrow 2000$, consistent with the claimed linear (L^1) scaling. SQL wall clock vs. ϵ_{fb} (at $L = 500$) fits exponent $\approx -2.06 / -2.05$, and COHERENT wall clock vs. ϵ_{fb} fits $\approx -1.03 / -1.04$, confirming the claimed ϵ_{fb}^{-2} (SQL) vs. ϵ_{fb}^{-1} (COHERENT) separation.

All ten checks (four ratios, two L -scaling segments, and four ϵ_{fb} -scaling segments) pass against the stated tolerances. The computation is scripted and provided with the source.

Exact-statevector simulation of FALQON convergence

We report exact-statevector simulations of FALQON on the critical transverse-field Ising model. Under a fixed step the algorithm reaches a useful minimum with a monotone Lyapunov descent that is stable across system size, confirming the descent mechanism at 4–10 qubits; the 60-qubit regime is beyond exact simulation, and the scaling there is an extrapolation from the mechanism verified here.

Protocol. Exact-statevector FALQON on the critical TFIM (open chain), $H_p = -J \sum_i Z_i Z_{i+1} - g \sum_i X_i$ with $J = g = 1$, driver $H_d = \sum_i X_i$, initial state $|+\rangle^{\otimes n}$, step $dt = 0.03$, $L = 500$ layers, feedback law $\beta_{k+1} = -\langle i[H_d, H_p] \rangle_k$ ($\beta_1 = 0$). The approximation ratio is $r = (E_{\text{max}} - E_{\text{final}})/(E_{\text{max}} - E_{\text{min}})$, equal to 1 at the exact ground state.

Result (swept $n = 4, 6, 8, 10$). The descent is monotone: the worst per-layer energy increase relative to $|E_{\min}|$ is at most 2.2×10^{-7} at every swept size (tolerance 10^{-3}). It reaches a useful minimum, $r = 0.99986/0.99944/0.99946/0.99949$ at $n = 4/6/8/10$ (floor 0.90), and it is size-stable — the ratio spread across sizes is 4.2×10^{-4} , with no degradation as n grows.

Boundary probe ($n = 12$, same fixed dt). At the same fixed step the $n = 12$ probe enters mild instability,

$r = 0.959$ with a worst relative per-layer increase of 1.1×10^{-2} : the fixed-step stability edge near $dt \sim 1/\|H\|$.

Extrapolation to 60 qubits. The 4–10 qubit simulations confirm the monotone Lyapunov descent and its size-stability. The 60-qubit, 500-layer regime of the main text lies beyond exact simulation; convergence there is an extrapolation from the mechanism verified here, not a directly simulated result. All three gated checks (monotone descent, useful minimum, size-stability) pass at the declared tolerances over $n = 4$ –10.

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- [1] A. G. Fowler, M. Mariantoni, J. M. Martinis, and A. N. Cleland, Surface codes: Towards practical large-scale quantum computation, *Physical Review A* **86**, 032324 (2012), arXiv:1208.0928.
 - [2] D. Litinski, A game of surface codes: Large-scale quantum computing with lattice surgery, *Quantum* **3**, 128 (2019), arXiv:1808.02892.
 - [3] N. J. Ross and P. Selinger, Optimal ancilla-free clifford+t approximation of z-rotations, *Quantum Information and Computation* **16**, 901 (2016), arXiv:1403.2975.
 - [4] A. B. Magann, K. M. Rudinger, M. D. Grace, and M. Sarovar, Feedback-based quantum optimization, *Physical Review Letters* **129**, 250502 (2022), arXiv:2103.08619.
 - [5] A. B. Magann, K. M. Rudinger, M. D. Grace, and M. Sarovar, Lyapunov-control-inspired strategies for quantum combinatorial optimization, *Physical Review A* **106**, 062414 (2022), arXiv:2108.05945.
 - [6] I. D. Kivlichan, Y. Su, D. W. Berry, R. Babbush, *et al.*, Improved fault-tolerant quantum simulation of condensed-phase correlated electrons via trotterization, *PRX Quantum* **1**, 020312 (2020), arXiv:1902.10673.
 - [7] A. M. Childs, Y. Su, M. C. Tran, N. Wiebe, and S. Zhu, Theory of trotter error with commutator scaling, *Physical Review X* **11**, 011020 (2021), arXiv:1912.08854.
 - [8] A. M. Childs and N. Wiebe, Hamiltonian simulation using linear combinations of unitary operations, *Quantum Information and Computation* **12**, 901 (2012), arXiv:1202.5822.
 - [9] R. Babbush, C. Gidney, D. W. Berry, N. Wiebe, J. McClean, A. Paler, A. Fowler, and H. Neven, Encoding electronic spectra in quantum circuits with linear t complexity, *Physical Review X* **8**, 041015 (2018), arXiv:1805.03912.
 - [10] E. T. Campbell, Early fault-tolerant simulations of the hubbard model, *Quantum Science and Technology* **7**, 015007 (2022), arXiv:2012.09238.
 - [11] G. H. Low, V. Kliuchnikov, and N. Wiebe, Well-conditioned multiproduct hamiltonian simulation, arXiv preprint (2019), arXiv:1907.11679.